

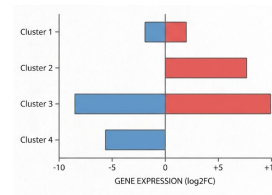
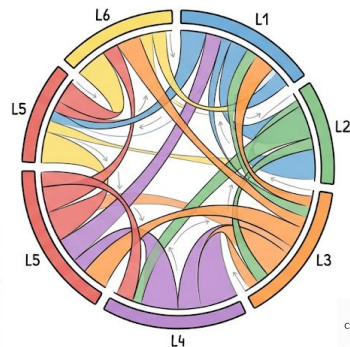
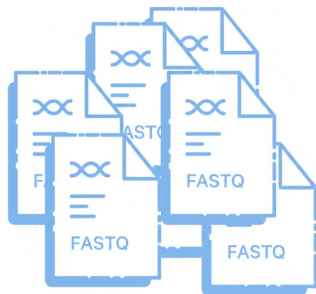
Hands-on Tour of Multi-Sample Single Cell Analysis

CG001730 RevA

Primary goal



Demystify the data analysis process with a hands-on, active learning approach.



Target audience



Those who have or will soon have 10x scRNA-seq data and have limited to no bioinformatics experience

Agenda

9:00 AM - 12 PM	Raw data processing and QA	• Sign in & Introduction • Overview of scRNA-seq Data Analysis • 10x Genomics Cloud Analysis: single sample • Quality Assessment • Loupe Browser: Quality control
12 PM - 1 PM	Lunch	
1 PM - 4:30 PM	Downstream analysis and visualization	• 10x Genomics Cloud Analysis: multi-sample integration • Loupe Browser: Cell type annotation and rapid data exploration • Introduction to community developed tools • Downstream analysis with community tools • Wrap up and optional office hour with the 10x team

Analysis journey: orientation

Experimental Design

Analysis begins with design. Choose assay types and define controls & replicates to set your course.

Lab Workflow

Use analysis-aware execution to avoid road-blocking technical artifacts.

Raw Data Processing

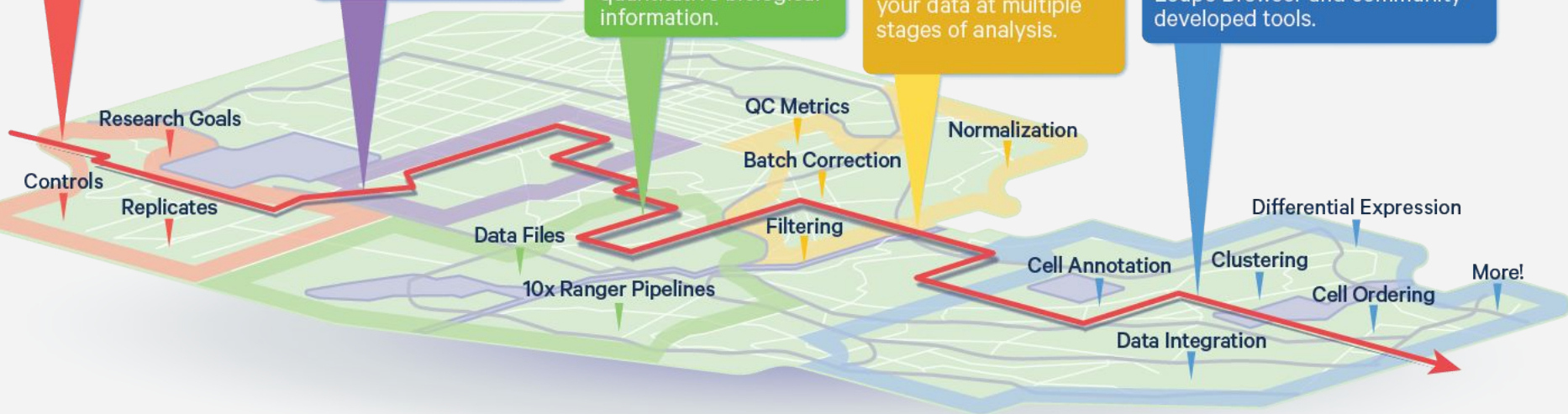
Transform your sequencing data into quantitative biological information.

Quality Control

Assess the quality of your data at multiple stages of analysis.

Visualization & Interpretation

Arrive at biological insights and address research questions with Loupe Browser and community-developed tools.



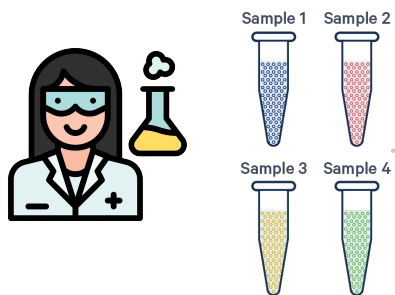
Introduction to the dataset: analysis begins at design

Hands on tour of the single cell analysis journey

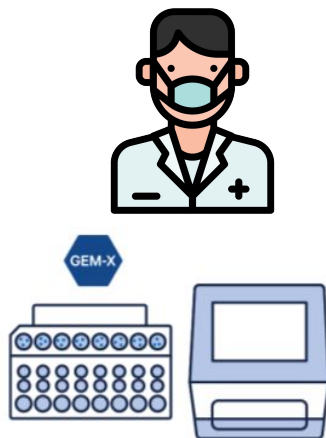
Outline

- Analysis begins at design
- Purpose of the experiment
- The data we are using today
- Analysis plan

Analysis begins at design



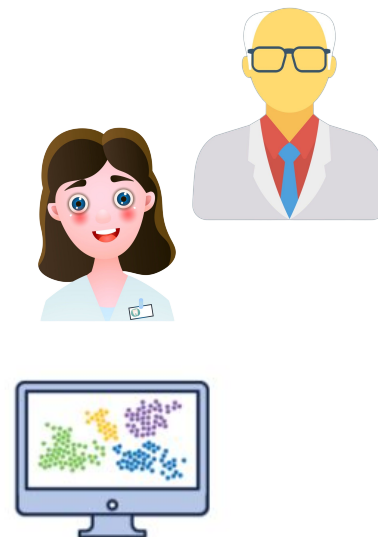
How many samples,
conditions do I need?



How many cells should
I target?



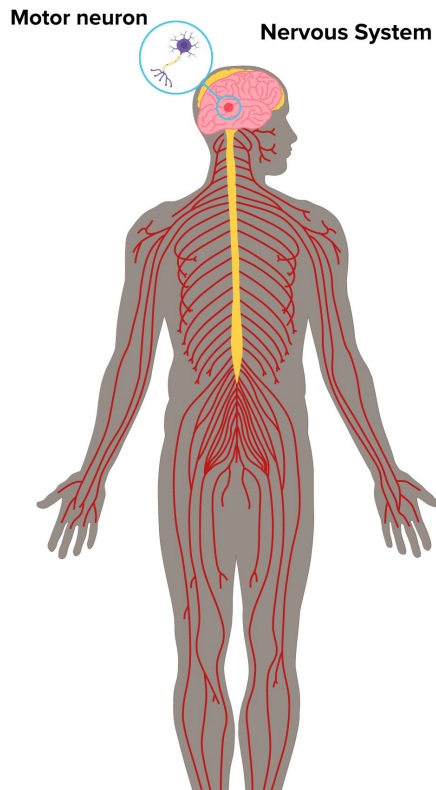
What sequencing
depth should I target?



What analysis tools
should I use?

The design will determine what research questions you can answer.

Amyotrophic Lateral Sclerosis (ALS)



**Symptoms
manifest in distal
limbs and spread**

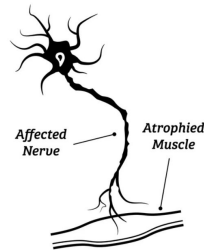
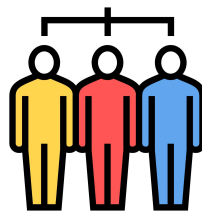


**Ultimately leads to
paralysis and death**

**Affects more than
200,000 people
worldwide**



**10% of cases are inherited
while 90% of cases are
sporadic**



**ALS kills motor
neurons and
weakens muscles**



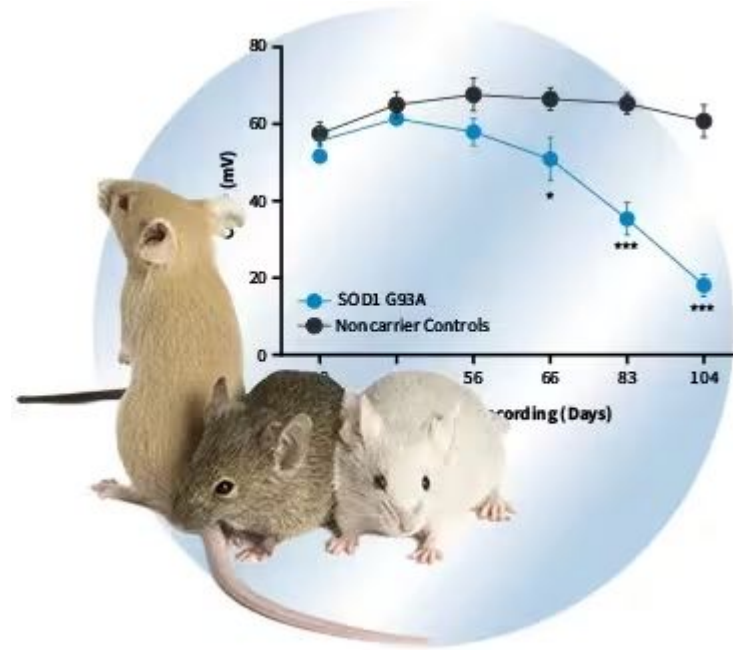
**There is no clear cause
or cure**

ALS mouse model

$SOD1^{G93A}$ mutation leads to toxic protein aggregation and motor neuron death

SOD1 mutations are one of the most common monogenic causes of familial ALS.

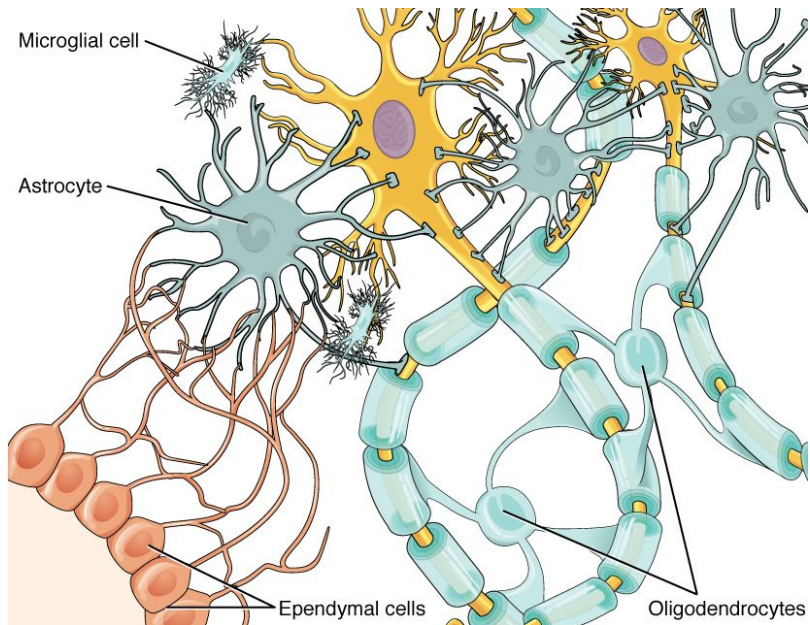
SOD1 mutant mice recapitulate the hallmarks of ALS including motor neuron degeneration and physical symptoms



<https://www.jax.org/jax-mice-and-services/preclinical-research-services/neurobiology-services/amyotrophic-lateral-sclerosis-efficacy-studies>

Single cell analysis to study complex disease

While the disease is driven by motor neuron death, other cell types contribute



<https://courses.lumenlearning.com/suny-ap1/chapter/nervous-tissue-2/>

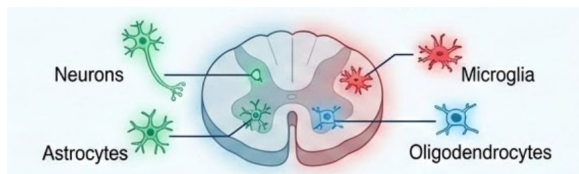
How non-neuronal cell types in the spinal cord contribute to disease remains poorly understood.

Single-cell sequencing offers a unique capability to deconvolute these complex, cell-type-specific interactions in vivo.

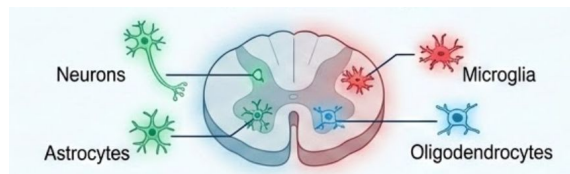
Purpose of the experiment

Our research questions that will drive the analysis:

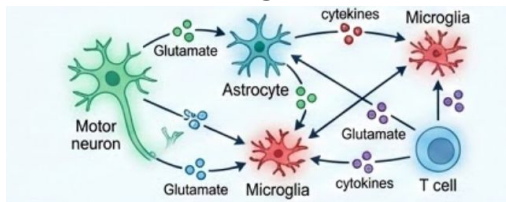
What cell types exist in our samples?



How do cell types change in ALS?



How do cell-cell interactions change in ALS?



Can we identify novel therapeutic targets with our results?



Experimental design

Goal: study molecular changes in motor neurons and their supporting glial cells in ALS

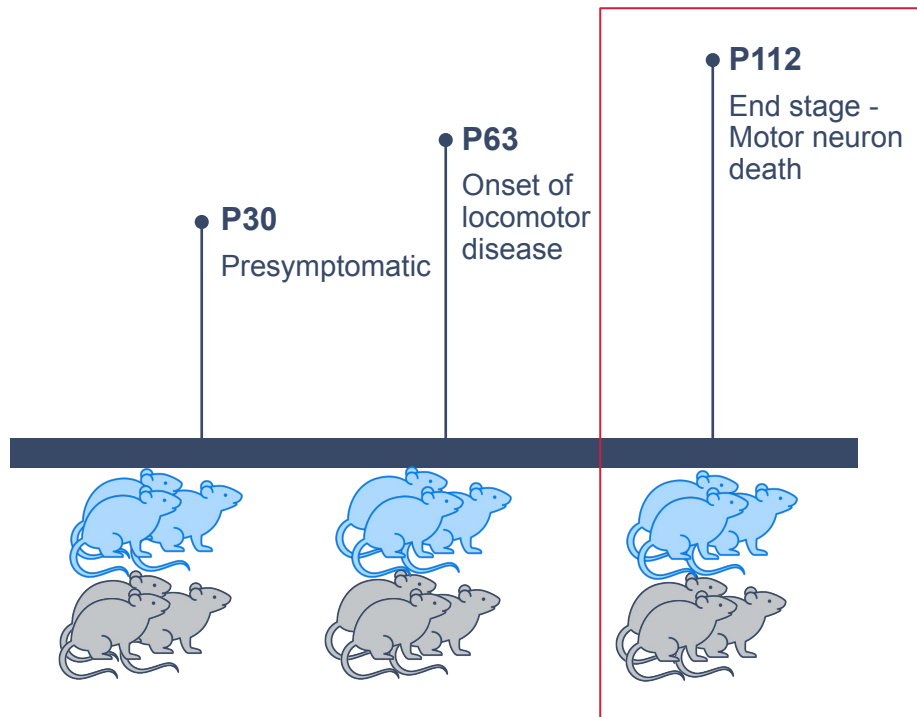
Lumbar spinal cords → GEM-X 3' v4



SOD1-G93A mutant carrier

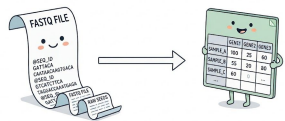


Wild-type

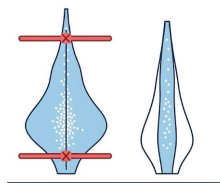


Common analysis flow for single cell data

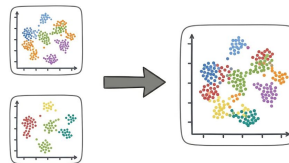
**Cell Ranger:
read processing &
secondary
analysis**



QC & data filtering



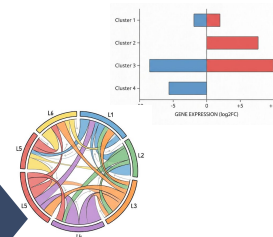
Data integration



**Visualization and
exploration**

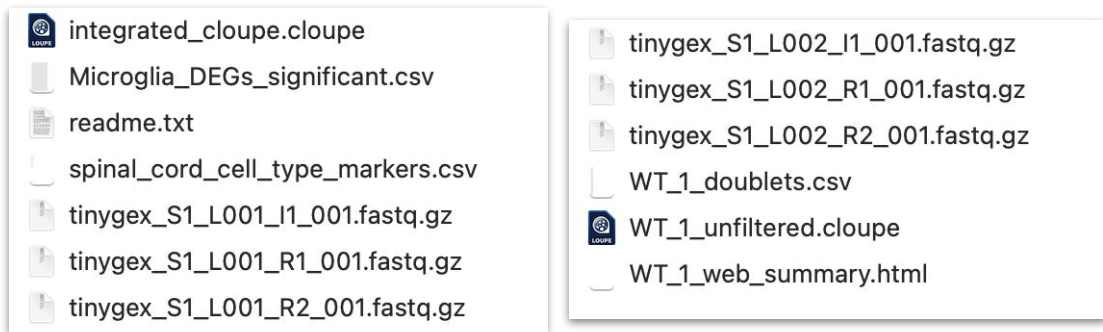


**Multisample
comparison and
more**



Prerequisites/Preparations

- Download and install latest [Loupe Browser](#)
- Download and unzip `sc-workshop-multi-sample-v1.0.zip`:



- Have a Google account (to be able to use CoLab)

Introduction to scRNA-seq data analysis

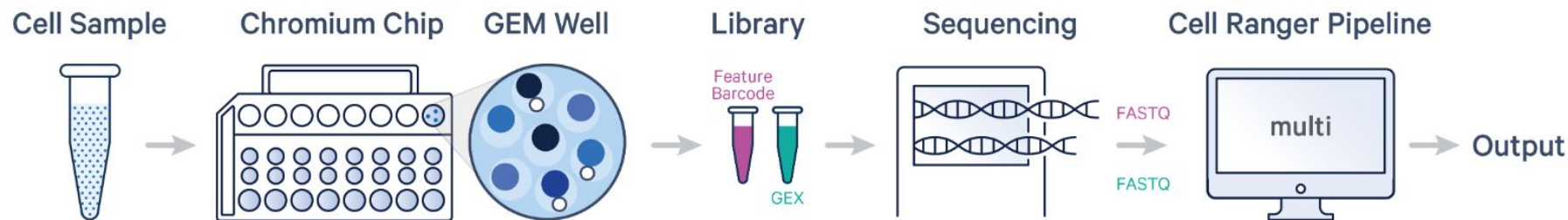
Hands on tour of the single cell analysis journey

Outline

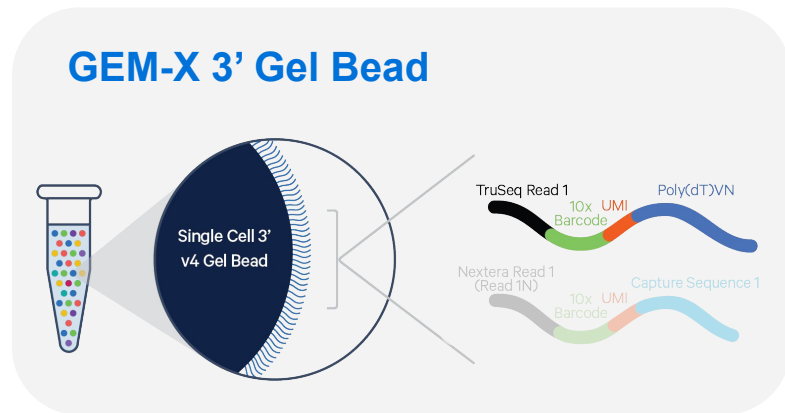
- Workflow overview
- Read structure
- Common analysis flow

Common scRNA-seq workflow

Laboratory and analysis workflow overview



Chromium Universal 3' GEM-X



Q: What identifies the cell of origin for a transcript?

i. TruSeq Read 1

22 nt Partial Illumina TruSeq Read 1 sequence

ii. 10x Barcode

16 nt 10x Barcode

~3.6 M defined barcode sequences

iii. UMI

12 nt Unique Molecular Identifier

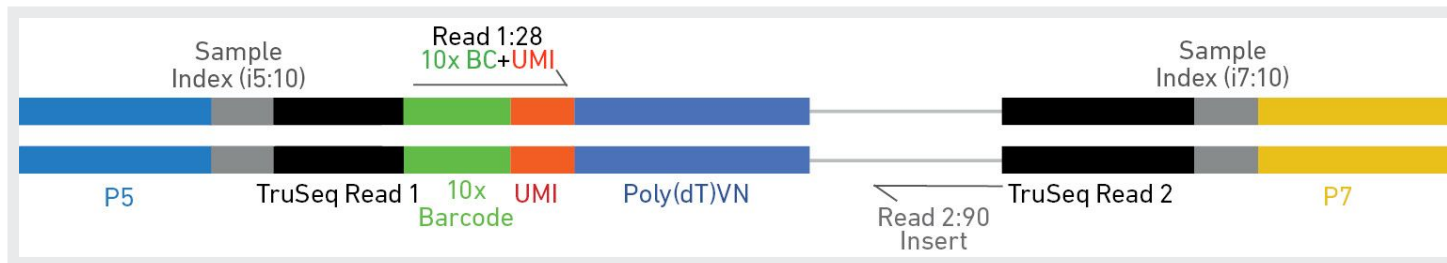
iv. Poly(dT)VN

30 nt Poly(dT) sequence

Enables capture of poly-adenylated mRNA molecules

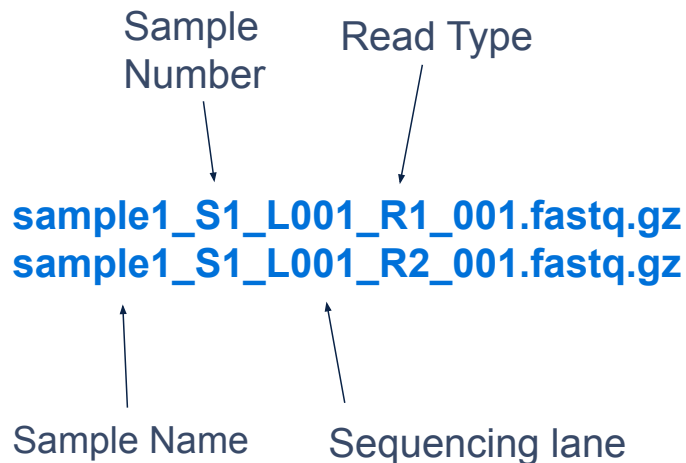
Read structure and sequencing

Single Cell Universal 3' Gene Expression library



	Read 1	i7 Index	i5 Index	Read 2
Purpose	10x Barcode & UMI	Sample Index	Sample Index	Transcript
Length	28	10	10	90

Anatomy of fastq file names

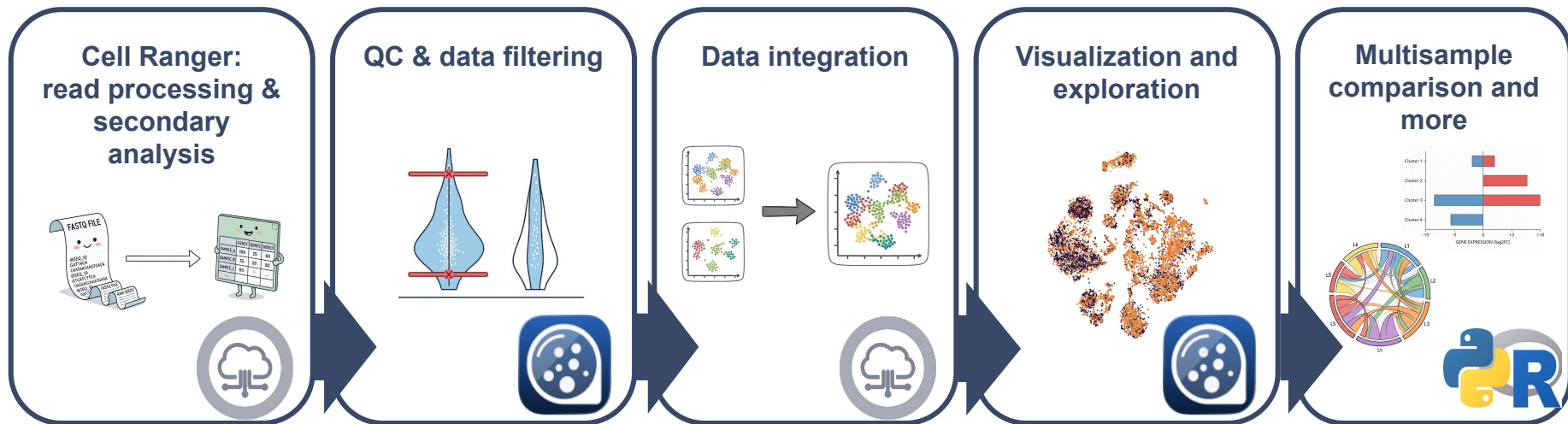


Read Type: Illumina sequencers typically output up to four types of reads

- R1: Read 1 - Cell barcode + UMI
- R2: Read 2 - Transcript Information
- I1: Index 1 - Index Read 1 (i7)*
- I2: Index 2 - Index Read 2 (i5)*

*Index reads are not always outputted depending on sequencing facility and are not required for analysis

Common analysis flow for single cell data



What software tools will we use to perform each step?



10x Genomics Cloud Analysis: single sample

Hands on tour of the single cell analysis journey

Outline

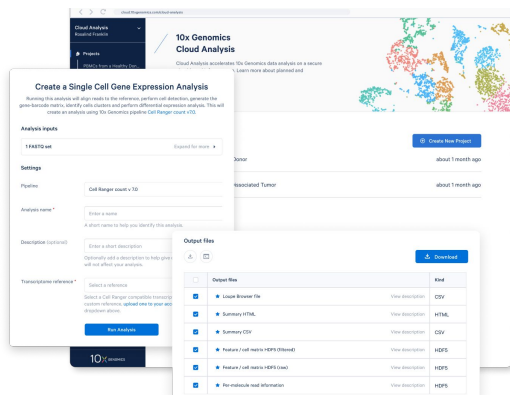
- Introduction to 10x Cloud Analysis
- Overview of Cell Ranger single sample analysis steps
- (Demo) Setup & process data on 10x Cloud

Introduction to 10x Cloud Analysis

10x Genomics Cloud Analysis

10xgenomics.com/cloud

Run Cell Ranger with a simple web interface



Access optimized runtimes on infrastructure tailored to 10x pipelines

Cloud Analysis

Optimized cloud clusters
512 cores | 4096 GB RAM

1 hr 20 mins

DIY cloud computing

Large instance on a public cloud service
64 cores | 488 GB RAM

5 hrs 4 mins

On-premises hardware

Minimum requirements for Cell Ranger
16 cores | 64 GB RAM

17 hrs 34 mins

*Time to process 30,000 cells using Cell Ranger multi v8.0
Typical results based on a time trial. Actual time may vary.

Free limits for all users to process compatible dataset

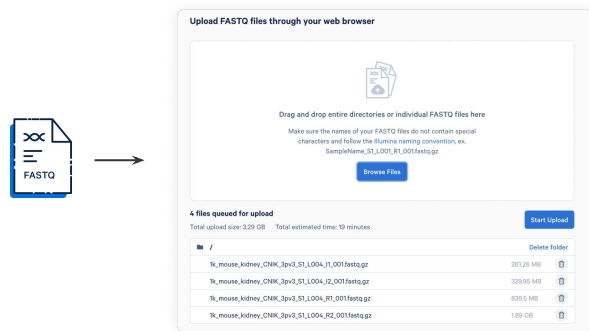
\$0
per sample

See [10x Genomics Cloud Terms of Use](#) for restrictions and details.
Learn more about [billing](#).

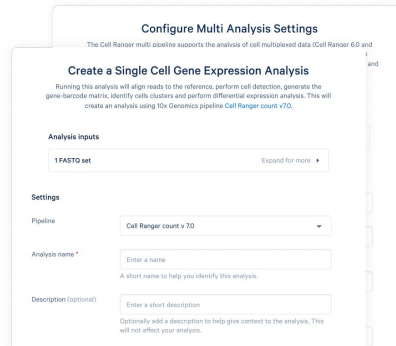
Using 10x Cloud: web interface or command line

1. Upload FASTQs

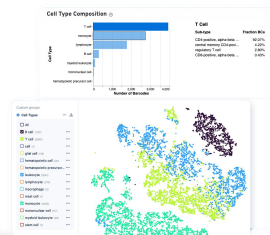
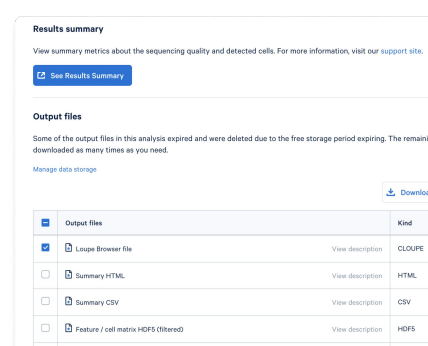
Intuitive browser-based workflows when you need guidance



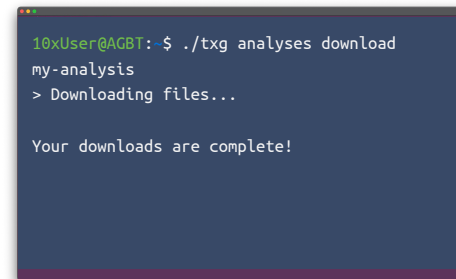
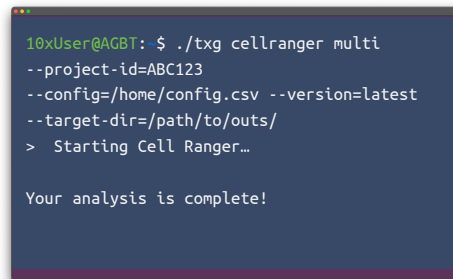
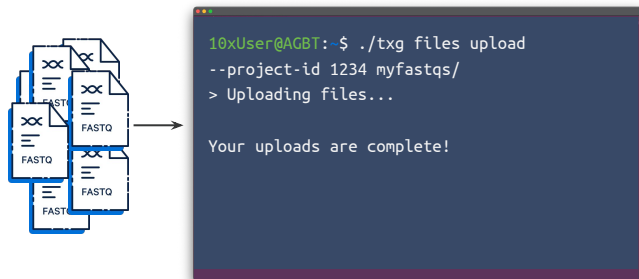
2. Start analyses



3. Download output



Scriptable input to 10x Cloud when you need to scale!



Overview of Cell Ranger single sample analysis steps

What is Cell Ranger?

- A suite of analysis pipelines that process Chromium Single Cell data
- Processing raw reads from a single GEM well: **multi** or **count**



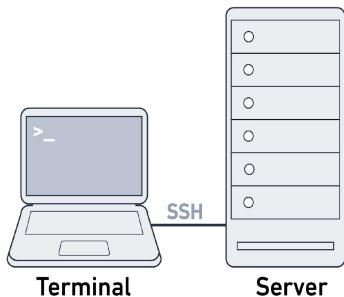
Running Cell Ranger



On 10x Cloud: Friendly user interface and simple data management

Required skill:

Understand the experimental design

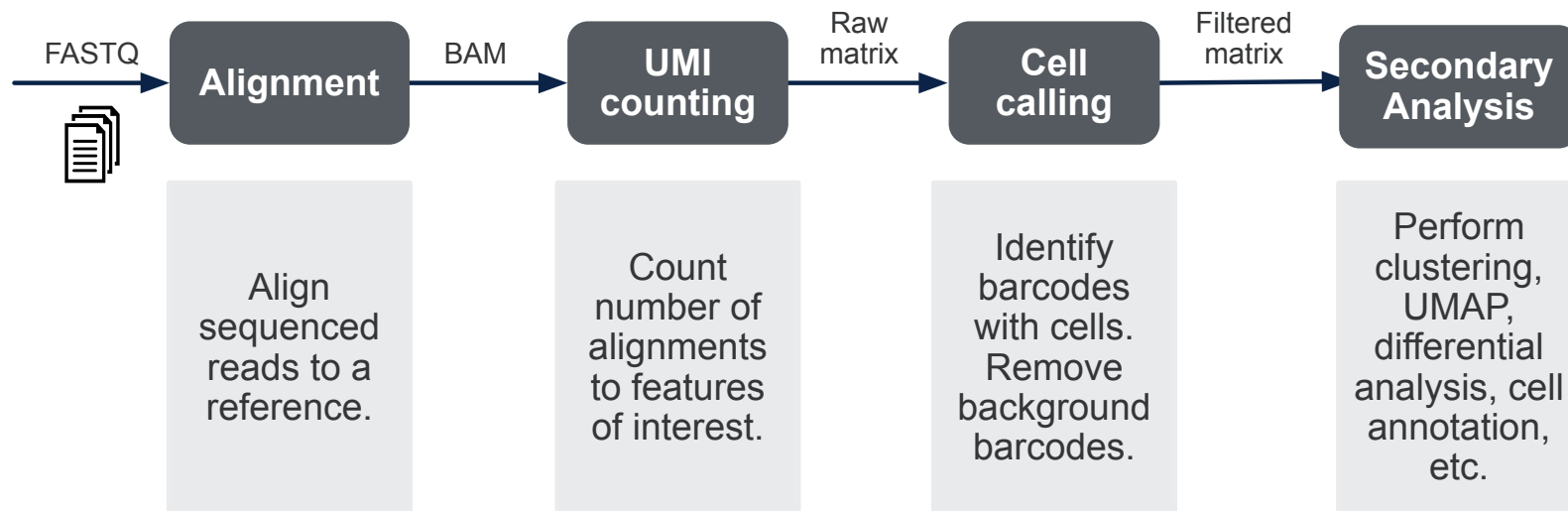


On a Linux system

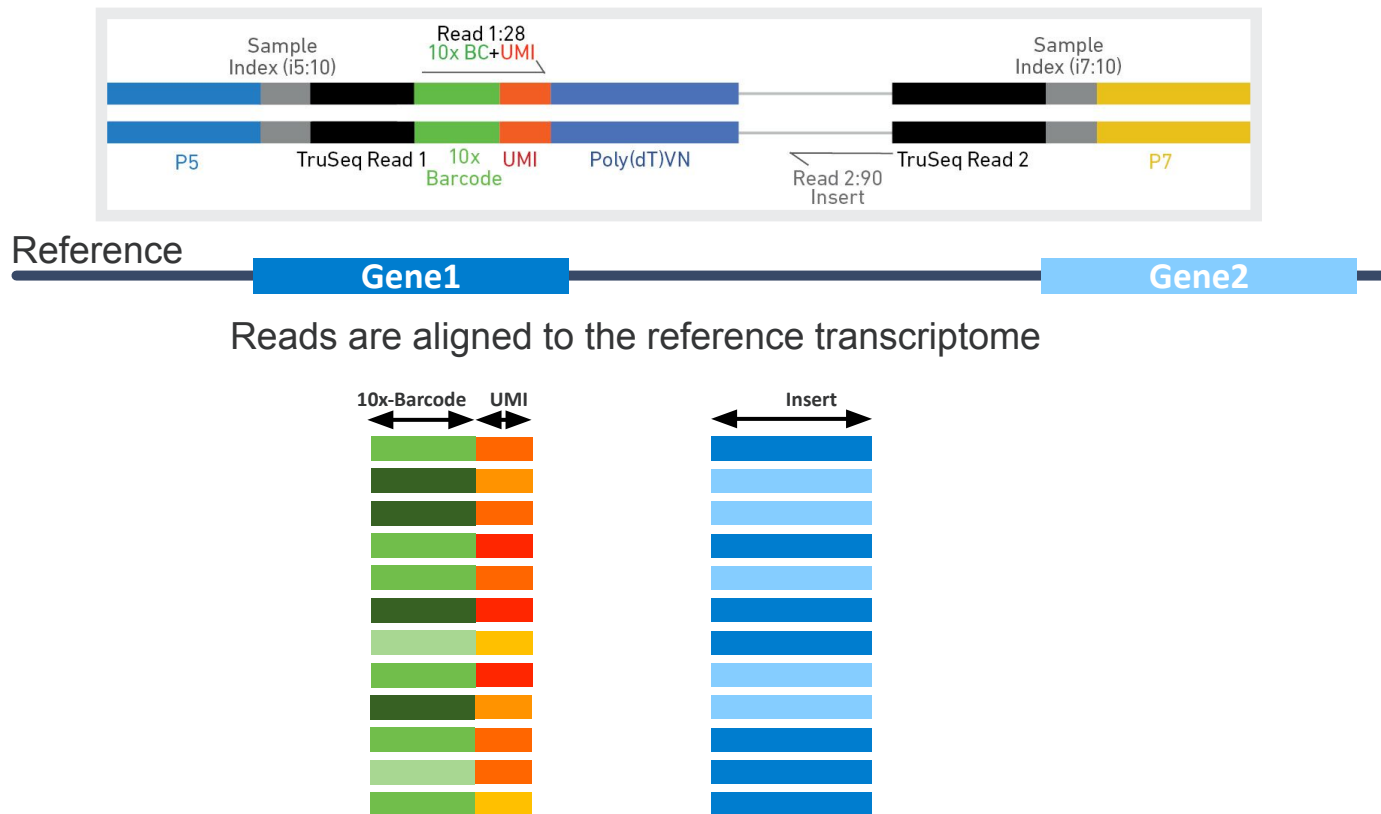
Required skills:

- Understand the experimental design
- Comfortable running command line in the Linux environment
- Familiar with organization's data management systems
- Know who to contact if there are server issues

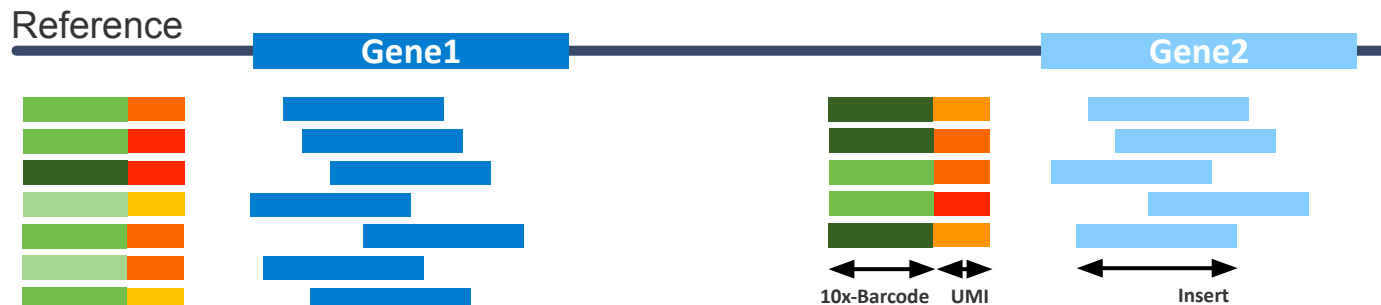
Cell Ranger single sample analysis steps



Cell Ranger analysis step: Alignment



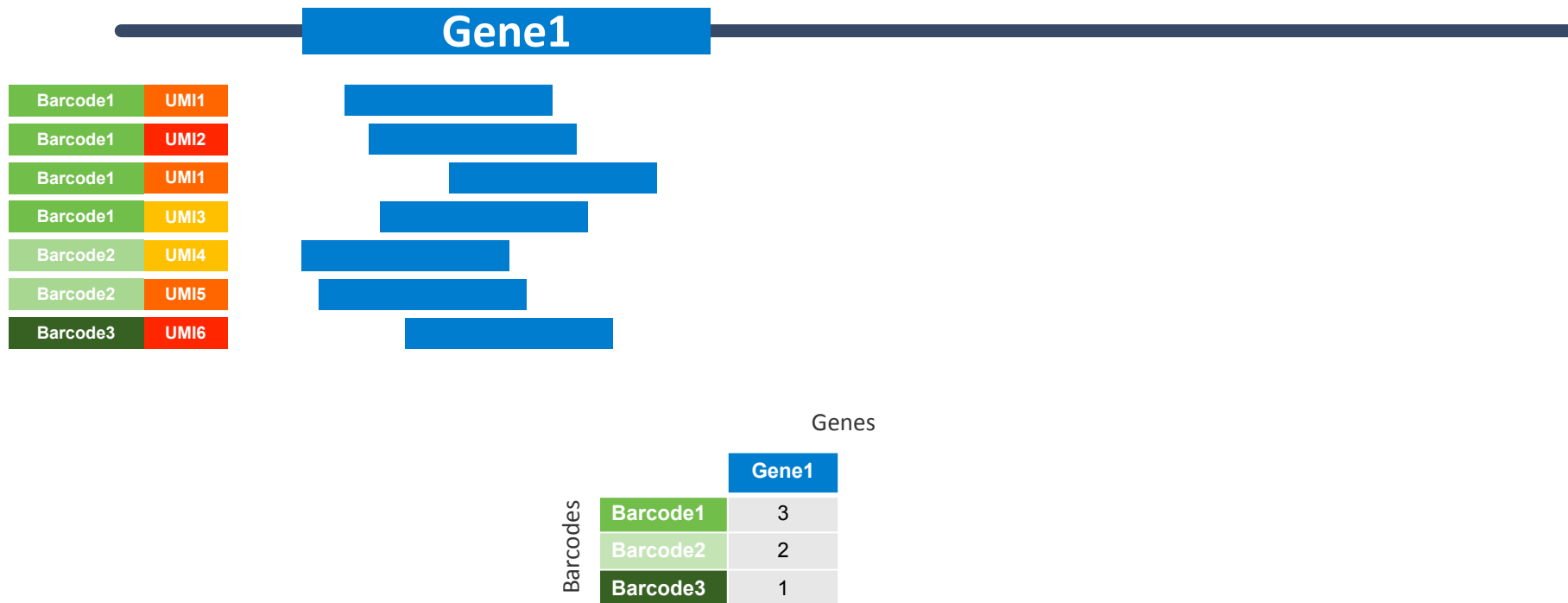
Cell Ranger analysis step: Alignment



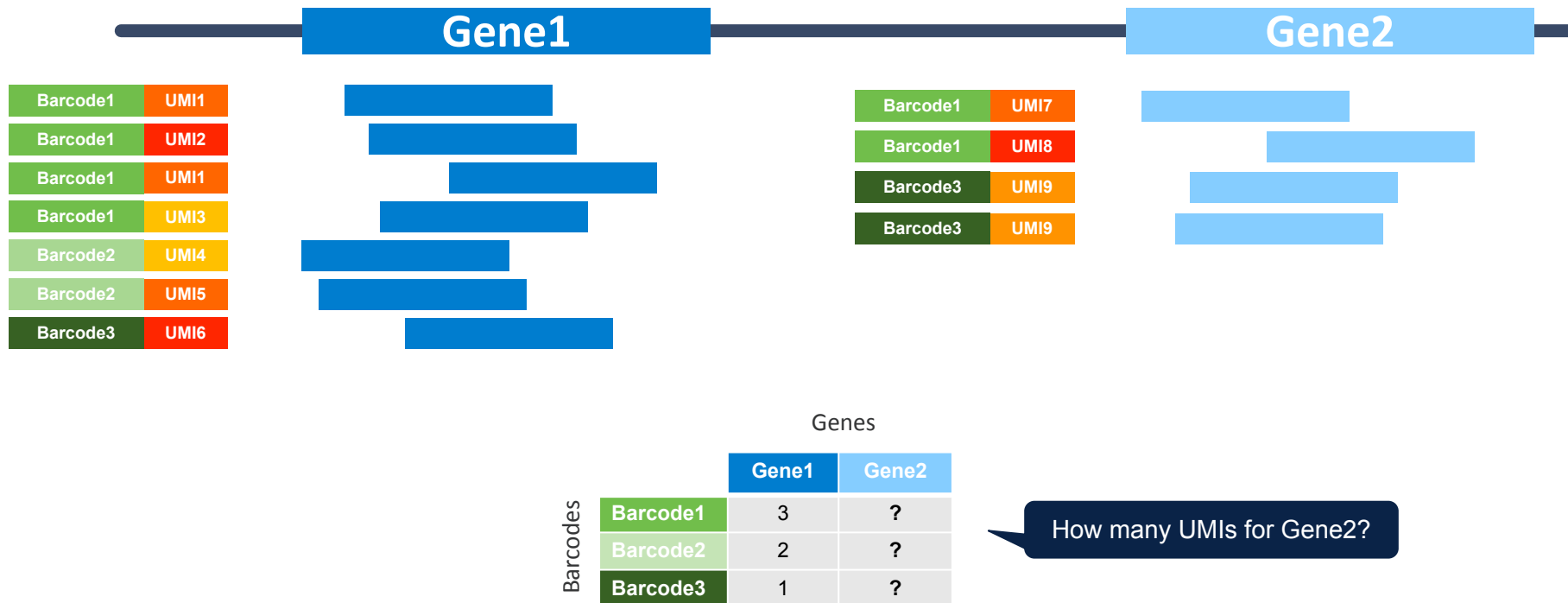
Alignment done via STAR (Spliced Transcripts Alignment to a Reference)

- Robust, open-source, junction-aware RNA-seq aligner
- Aligns reads to the genome and transcriptome simultaneously

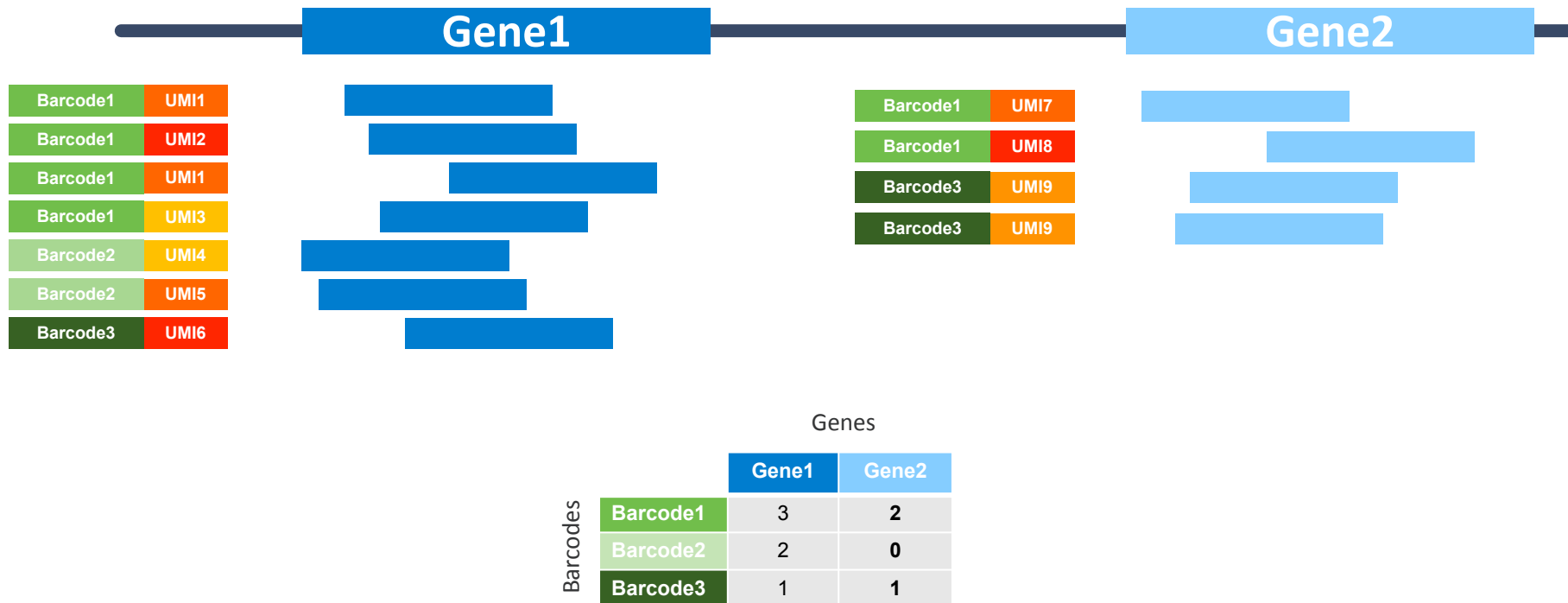
Cell Ranger analysis step: UMI counting



Cell Ranger analysis step: UMI counting

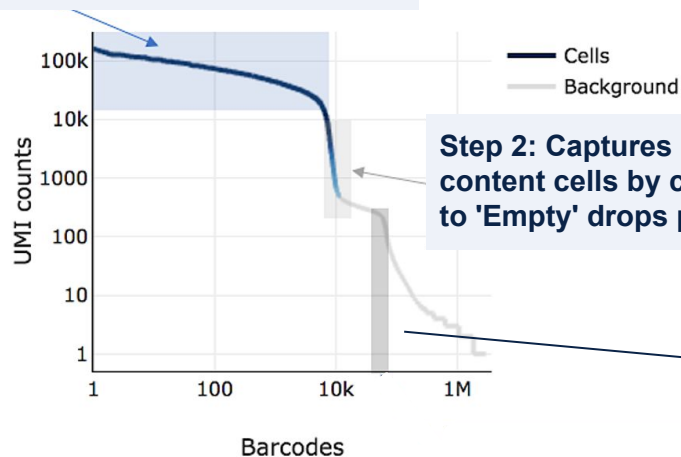


Cell Ranger analysis step: UMI counting

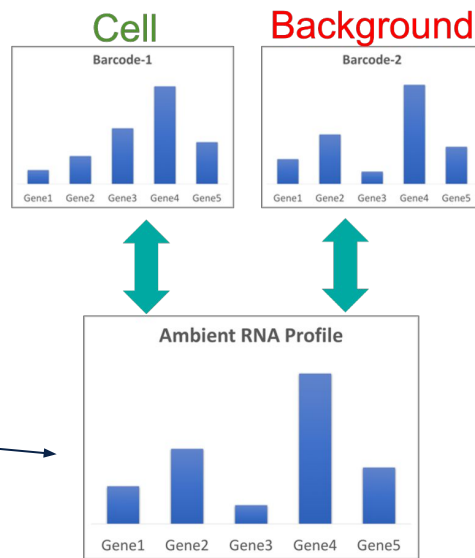


Cell Ranger analysis step: Call Cells

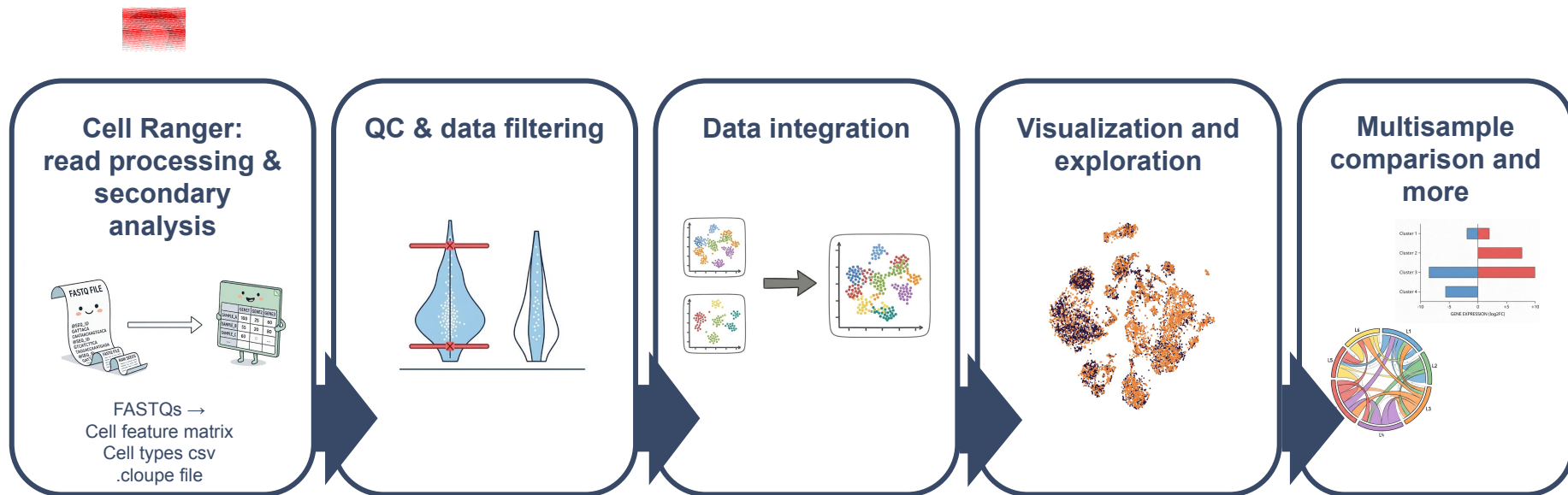
Step 1: identify the initial number of high RNA content cells based on UMI counts



Step 2: Captures low RNA content cells by comparing them to 'Empty' drops profile



Common analysis flow for single cell data



Setup & process data on 10x Cloud

Learning objective

We will use the 10x Cloud Analysis platform to process GEM-X 3' single-cell Gene Expression data.

We will practice the following:

1. Signing in and creating a project
2. Upload tiny FASTQ files for practise (presenter demo on how to use CLI)
3. Run analysis for ALS samples
4. Check output files

Datasets for hands-on exercise

Tiny FASTQ dataset

- Purpose:
 - Practice creating project
 - Practice uploading FASTQ in web
- Download link can be found in your welcome email or on the agenda page.
 - https://cf.10xgenomics.com/supp/cell-exp/workshop_tutorial/tiny_gex.zip

ALS dataset

- Purpose:
 - Run Gene Expression analysis
 - Filter individual sample .cloupe files
 - Run Data Integration using SCTransform
 - Outputs will be explored in Loupe Browser and community tools sections

Files are in the project transferred to you.

- Can also be accessed from [10x dataset page](#)

Interactive session

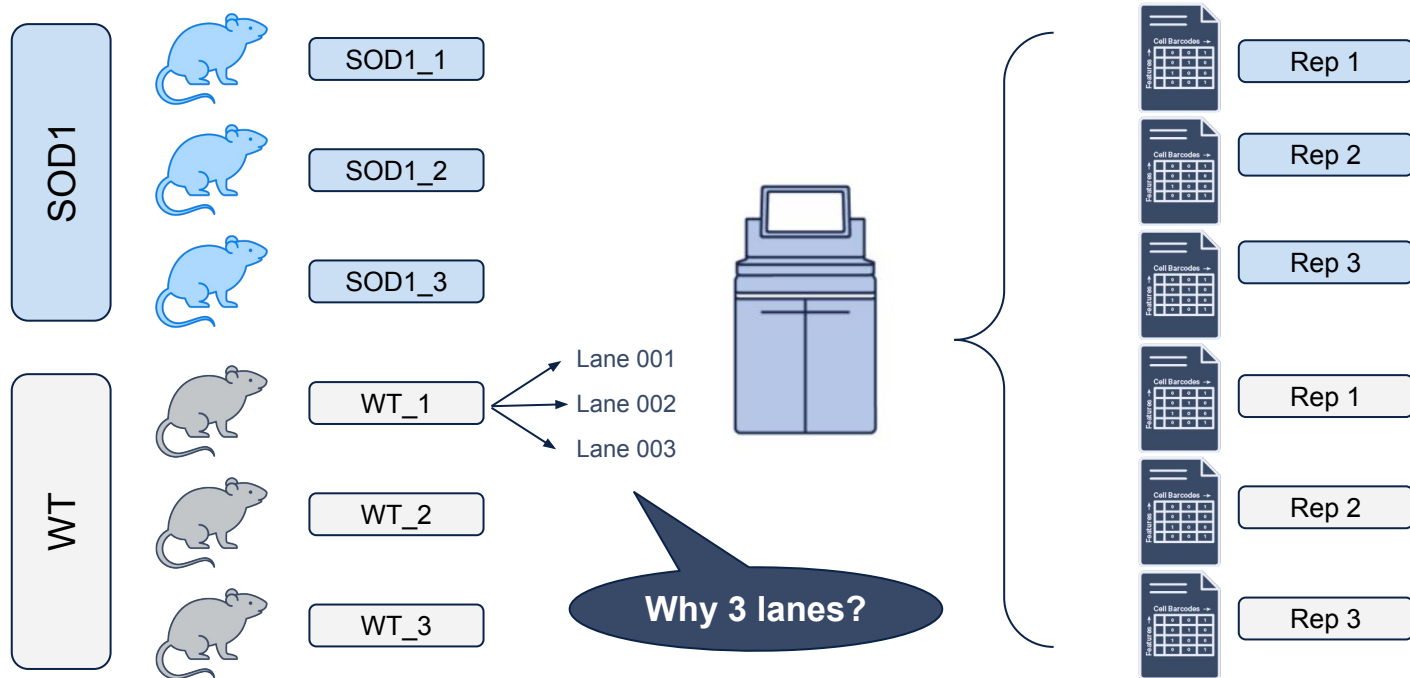
The dataset we are using today

Condition

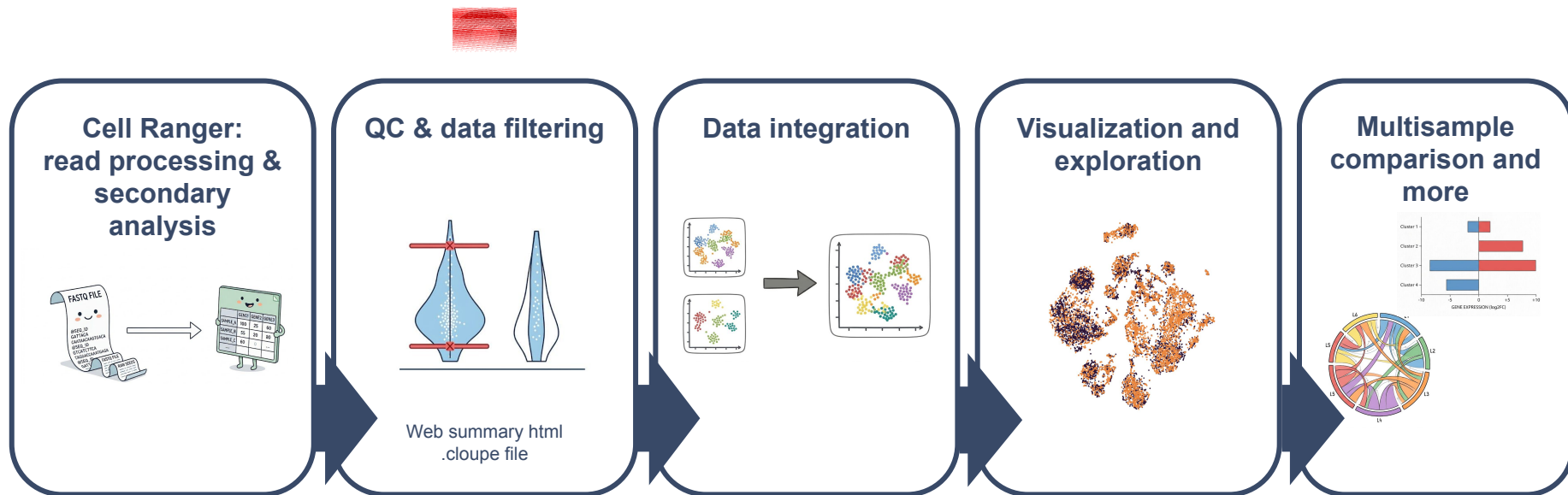
Sample

Sequencing data

Primary analysis



Common analysis flow for single cell data



Quality assessment: interpreting the Cell Ranger QC summary

Hands on tour of the single cell analysis journey

Outline

- Where to find the QC summary
- Hands-on walk through

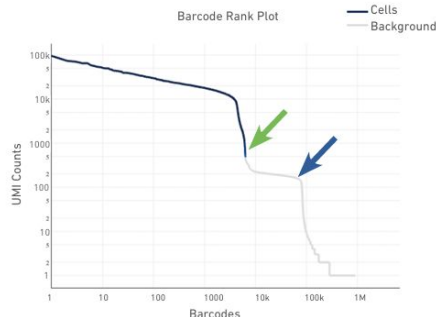
Interactive session

Demo: Example Barcode Rank Plots

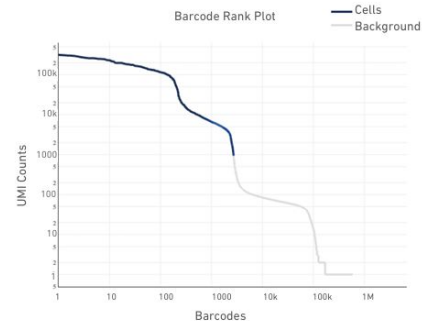
<https://www.10xgenomics.com/support/universal-three-prime-gene-expression/documentation/steps/sequencing/interpreting-cell-ranger-web-summary-files-for-single-cell-gene-expression-assays>

Examples

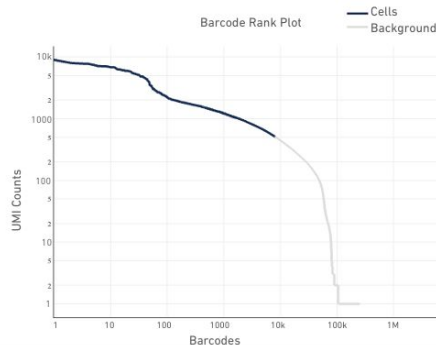
Typical Sample: A steep drop-off is indicative of good separation between the cell-associated barcodes and the barcodes associated with empty GEMs. An ideal Barcode Rank plot has a distinctive shape, which is referred to as a "cliff and knee". The blue-to-gray transition (green arrow) is referred to as the cliff; the solid gray is referred to as the knee (blue arrow).



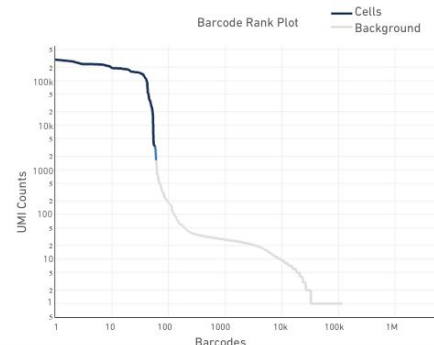
Heterogeneous Sample: Occasionally and based on sample type, there can be heterogeneous populations of cells in a sample that may result in a bimodal plot. In these situations, the cell-associated 10x Barcodes will have two "cliff and knee" distributions. However, there should still be clear separation between the barcodes called as 'cells' and barcodes called as 'background'.



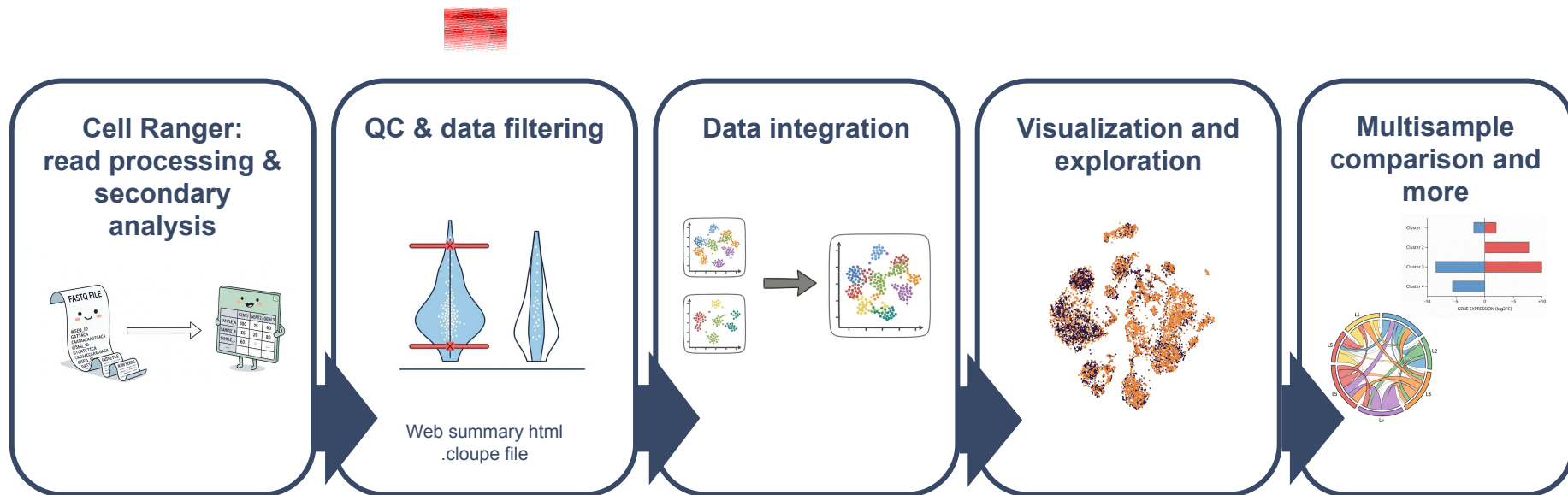
Compromised Sample: Round curve and lack of steep cliff may indicate low sample quality or loss of single-cell behavior. This can be due to a wetting failure, premature cell lysis, or low cell viability.



Compromised Sample: Defined cliff and knee, but the total number of barcodes detected may be lower than expected. This can be caused by a sample clog or inaccurate cell count.



Common analysis flow for single cell data



Loupe Browser: QC filtering

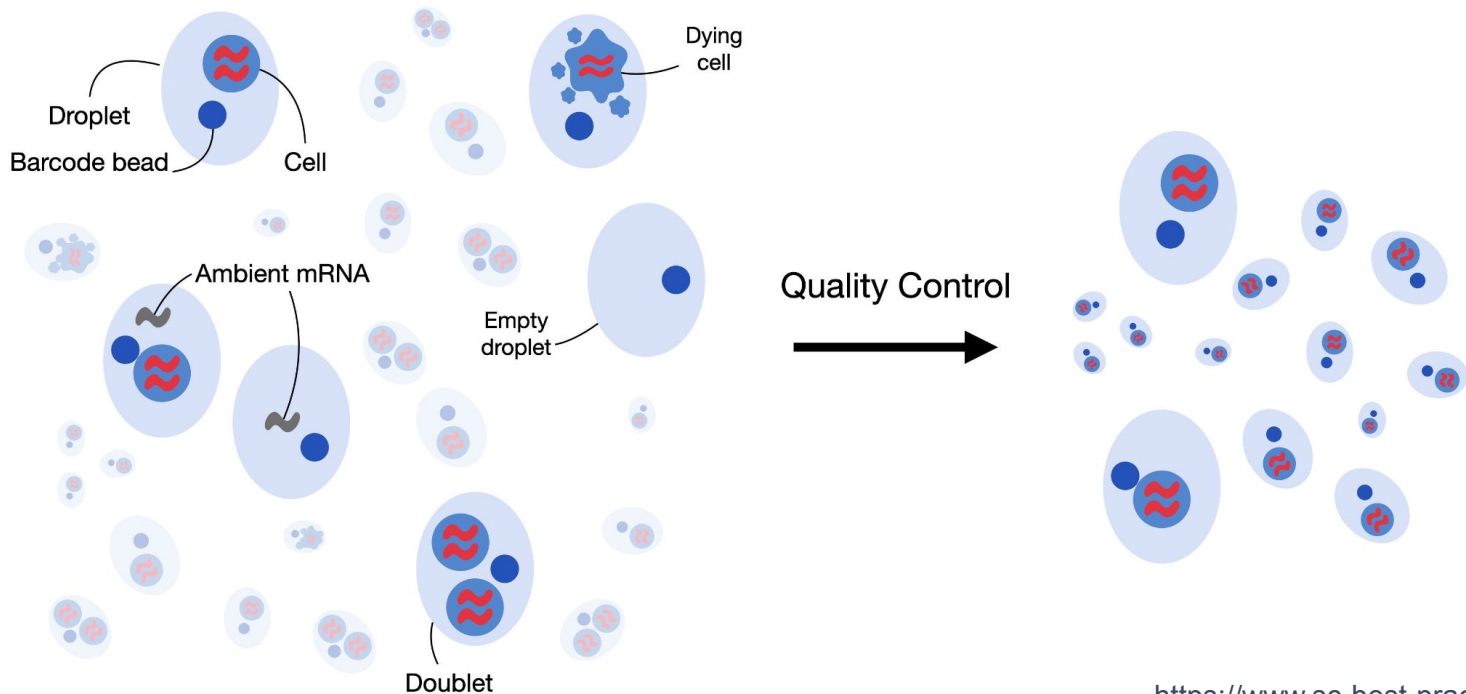
Hands on tour of the single cell analysis journey

Outline

- Quality control for single cell RNA-seq data
- Hands-on demonstration
- Multi-sample QC approach

Quality control for Single Cell RNA-seq data

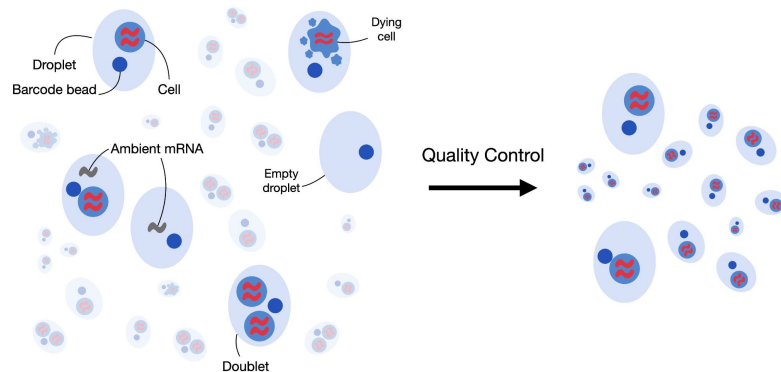
Assess quality and remove low-quality cells, cell-free RNA and doublets



<https://www.sc-best-practices.org/>

Quality control and assessment

Common QC steps for scRNA-seq



Removing empty droplets based on the expression profile

Filtering cells by percent of mitochondrial (mt) reads

Filtering cell barcodes by UMI counts or number of features

Doublet detection and removal

Removing ambient RNAs associated with barcodes

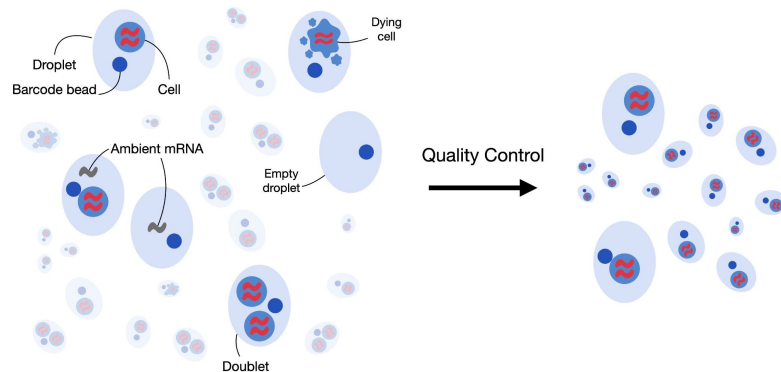
Cell Ranger, Loupe Browser

Community tools

Before performing any QC filters, it is important to first assess the overall quality and determine if additional QC filters are needed

Quality control and assessment

Common QC steps for scRNA-seq



Removing empty droplets based on the expression profile

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Cell Ranger, Loupe Browser

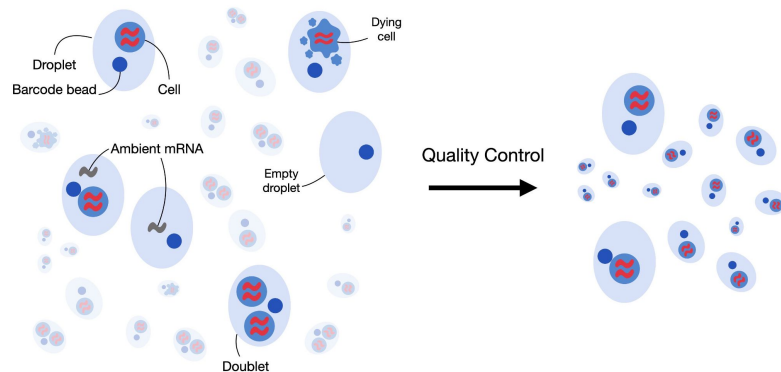
Community tools

[Introduction to ambient RNA correction analysis guide](#)

Tools: SoupX, CellBender, DecontX

Quality control and assessment

Common QC steps for scRNA-seq



Removing empty droplets based on the expression profile

Filtering cells by percent of mitochondrial (mt) reads

Filtering cell barcodes by UMI counts or number of features

Doublet detection and removal

Removing ambient RNAs associated with barcodes

Cell Ranger, Loupe Browser

Community tools

We used DoubletFinder to identify possible doublets in the data.

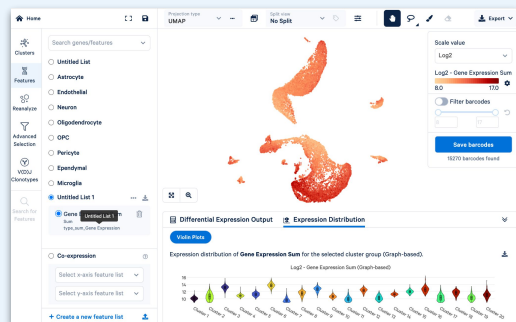
We will remove them in the Loupe Demo.

See accompanying tutorial (coming soon) for details.

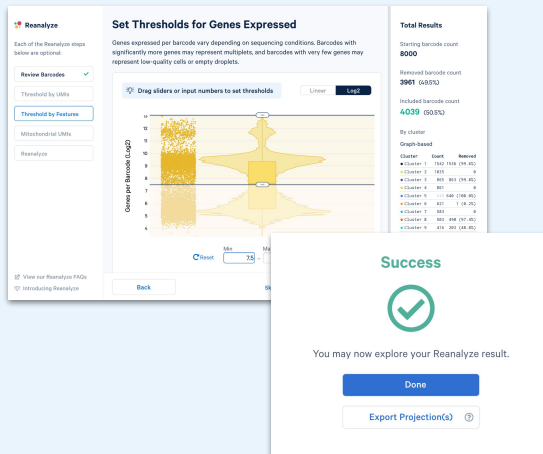
Loupe filtering overview

Main steps and data in Loupe filtering demo

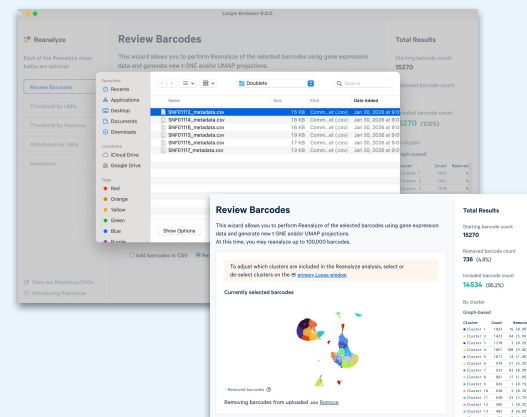
Identify low UMI, low quality clusters



Remove low quality cells



Remove doublets and re-cluster

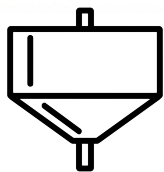
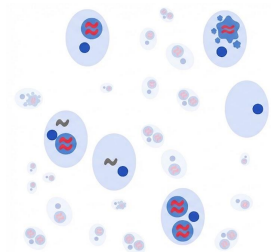


Interactive sessions

QC of multiple samples in the same study

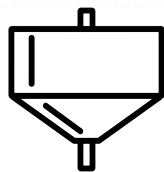
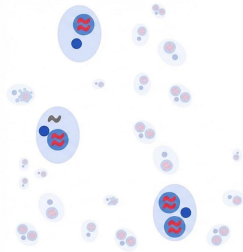
Data output from cellranger
(1 sample)

Repeat for each sample



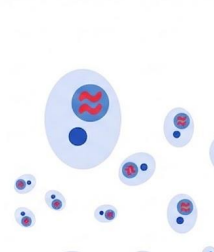
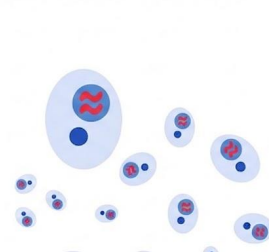
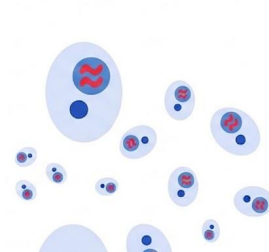
Remove empty
droplets and
dead/dying cells

Removed 5-10% low
UMI, low quality cells



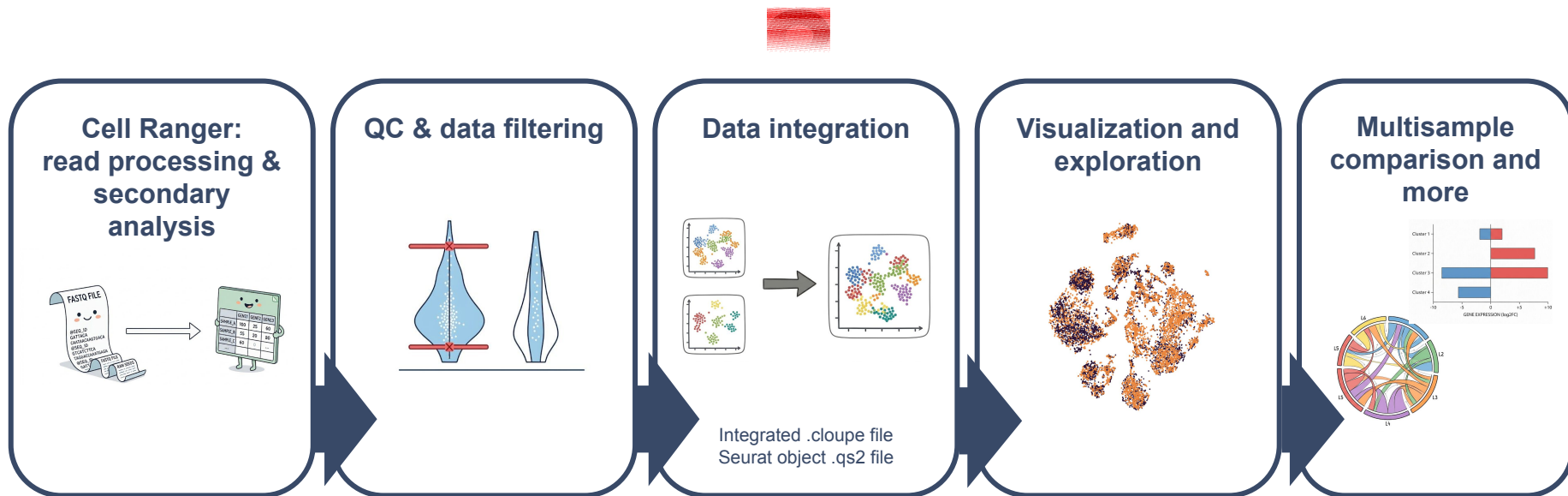
Remove doublets
and correct
ambient RNA

Removed 6%
doublets with
doubletFinder



Merge data
and analyze

Common analysis flow for single cell data



Agenda

9:00 AM - 12 PM	Raw data processing and QA	• Sign in & Introduction • Overview of scRNA-seq Data Analysis • 10x Genomics Cloud Analysis: single sample • Quality Assessment • Loupe Browser: Quality control
12 PM - 1 PM	Lunch	
1 PM - 4:30 PM	Downstream analysis and visualization	• 10x Genomics Cloud Analysis: multi-sample integration • Loupe Browser: Cell type annotation and rapid data exploration • Introduction to community developed tools • Downstream analysis with community tools • Wrap up and optional office hour with the 10x team

Multi-sample integration

Hands on tour of the single cell analysis journey

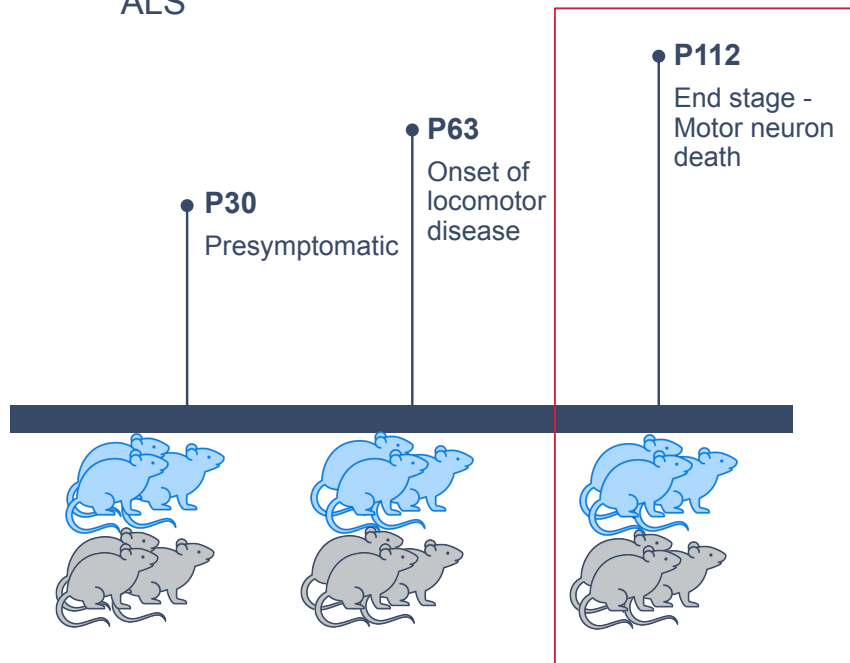
Outline

- Why multi-sample integration?
- 10x multi-sample integration tools
- Hands-on: setup multi-sample integration on 10x Cloud

Why multi-sample integration?

- Combine multiple samples into a shared analysis space so the same cell types align across samples
- It enables WT vs ALS comparison (cell types, states, markers) in one view.
- Reduces sample to sample technical differences so biology is easier to understand.

Goal: study molecular changes in motor neurons and their supporting glial cells in ALS



Multi-sample integration tools

Community developed tools

- Seurat's **SCTransform** for normalization (available on 10x Cloud, GEX only).
- **Harmony** for Batch correction (available on 10x Cloud, GEX only).

Cell Ranger

- **Aggr** pipeline to integrate `count/multi` results from individual samples (different modalities are supported).
- Chemistry batch correction algorithm in “aggr” works in PCA space and changes UMAP and cell clustering.

Introducing SCTransform

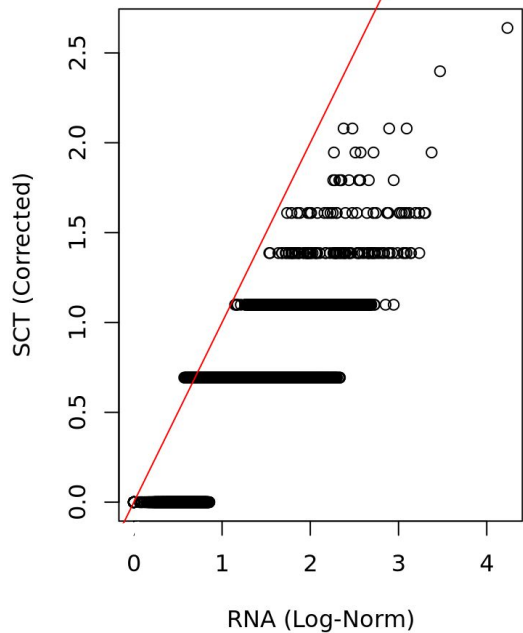
Model-based normalization at **per-sample level**.

Address technical noise: Removes the variations in gene counts due to differing sequencing depths among cells

Stabilize variance: Adjusts the data so that so we can properly compare the significance of changes, regardless of whether a gene is highly or lowly expressed.

[Seurat's SCTransform](#)

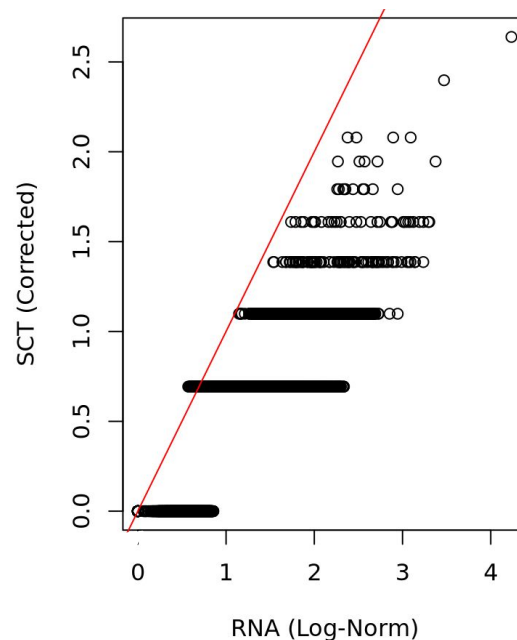
LogNorm vs SCTransform



Normalization methods: toy example

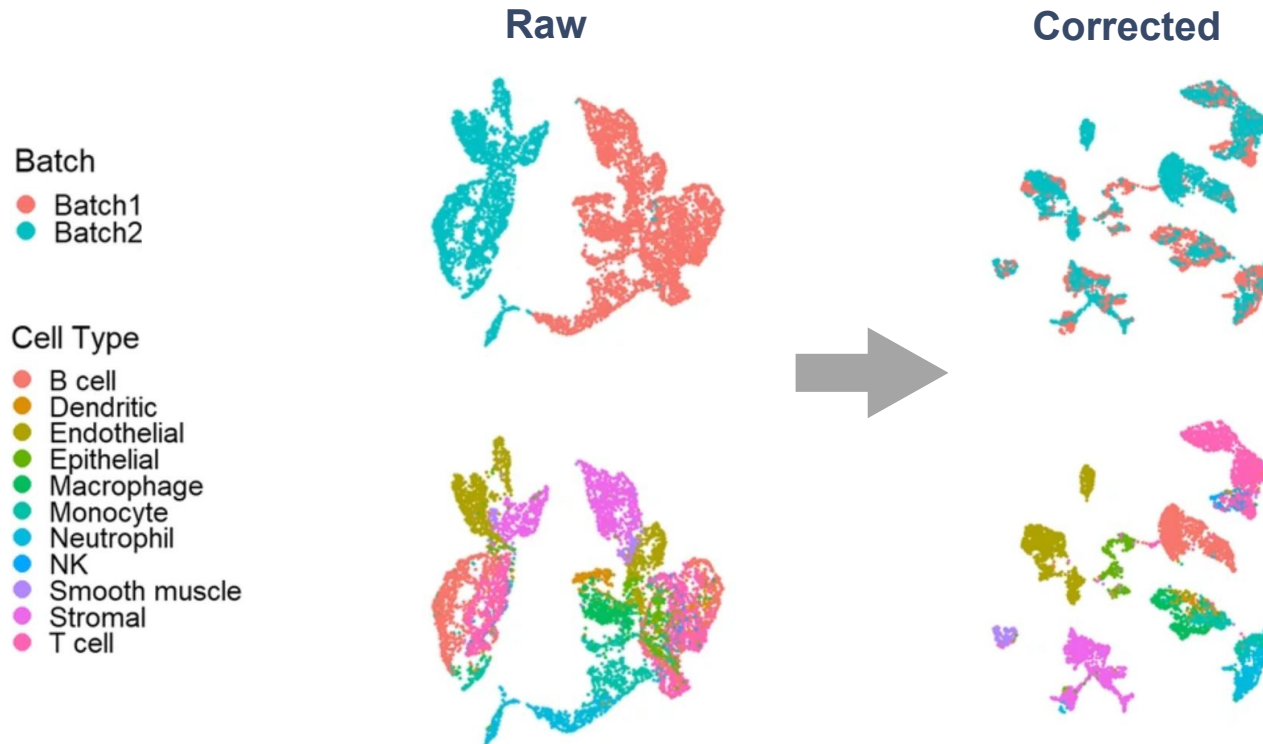
Cell	Total UMIs for cell	GeneX UMI counts	LogNorm intuition	SCT intuition
A	1000	1	Expressed	Maybe weak signal
B	10,000	1	Expressed	Likely noise
C	10,000	15	Expressed	Expressed
D	10,000	40	Highly expressed	Expressed

LogNorm vs SCTransform



Introducing batch effect correction

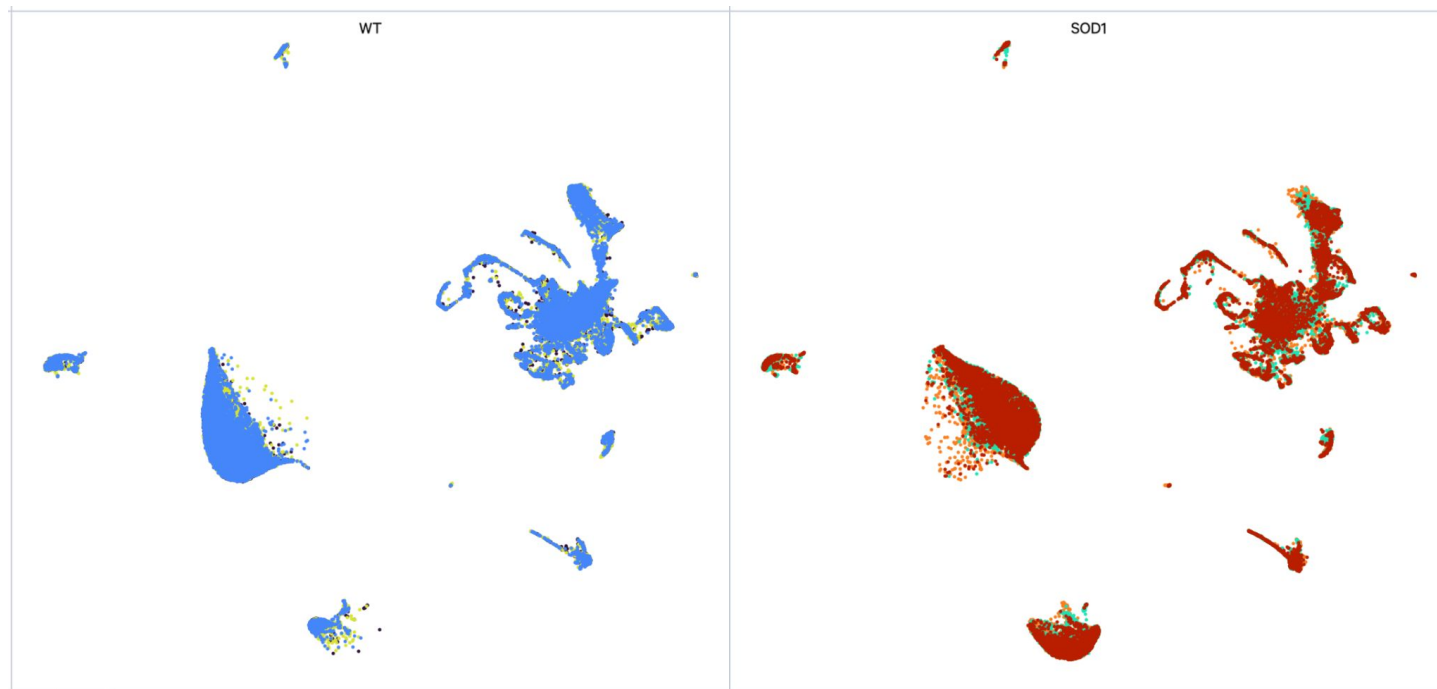
Effective methods remove technical variations while preserving cell type purity.



Genome biology 21.1 (2020): 1-32

Is batch correction needed for this dataset?

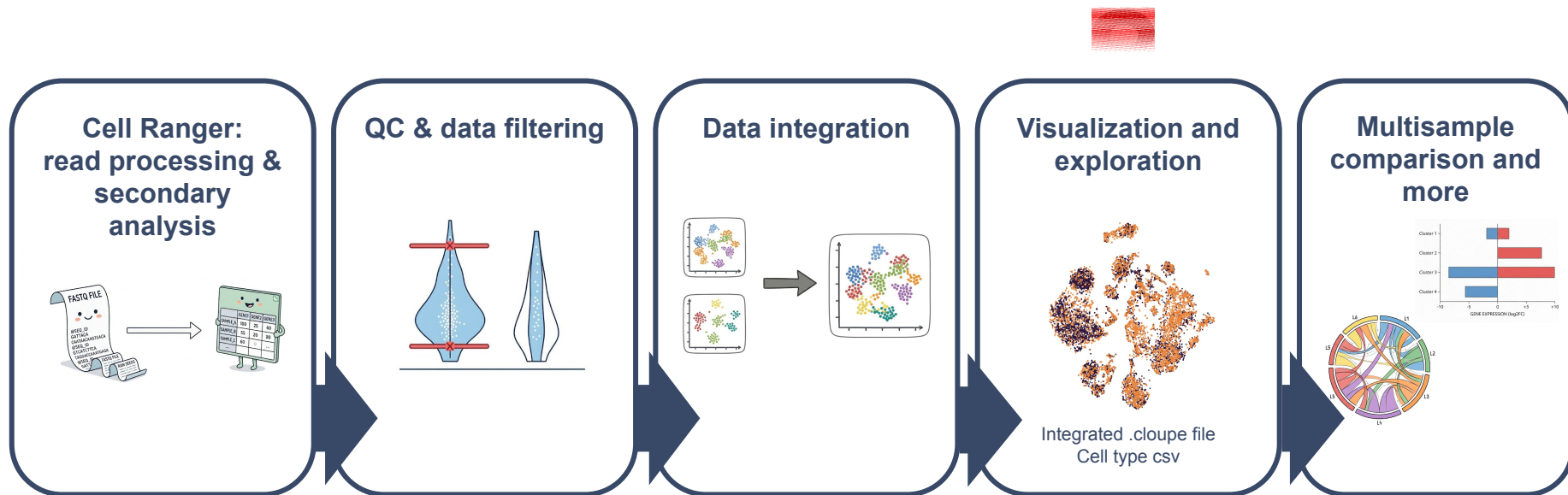
Perform batch correction ONLY if technical batch effects impact results



UMAPs from the .cloupe file separated by sample and condition to show no batch effect

Interactive sessions

Common analysis flow for single cell data



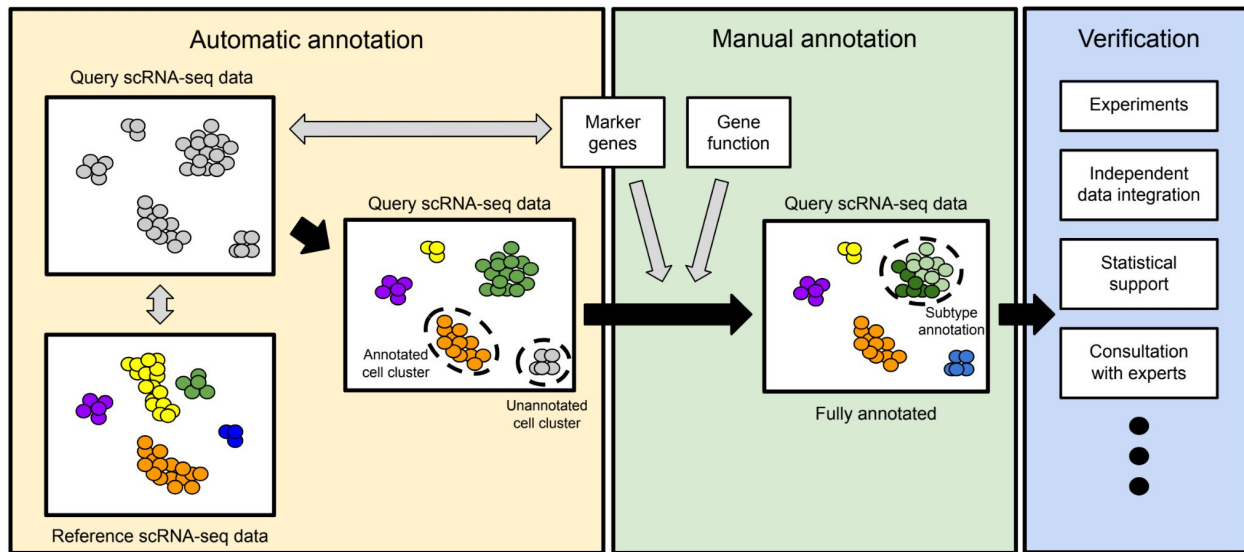
Loupe Browser: cell type annotation and initial data exploration

Hands on tour of the single cell analysis journey

Outline

- Introduction to cell type annotation
- Hands-on annotation in Loupe Browser

Introduction to cell type annotation



<https://www.nature.com/articles/s41596-021-00534-0>

Cell annotation

Introduction

Jun 28, 2023

Web Resources for Cell Type Annotation

The advantage of visualizing cell populations at single-cell resolutions has introduced us to the challenges of annotating cells. A growing number of annotation databases and tools are available to aid us in the process. In this article, we provide general guidance for a selected few to assist you with your research.

Cell annotation

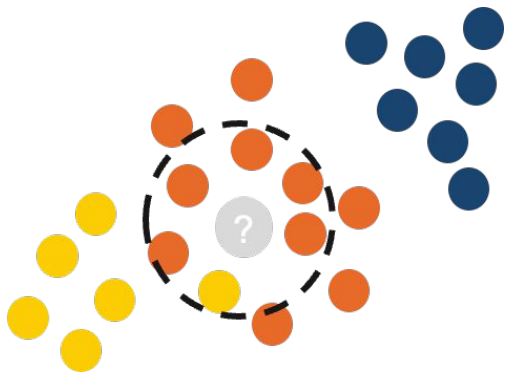
Tutorial

Jan 25, 2024

Automated cell type annotation from R to Loupe using LoupeR

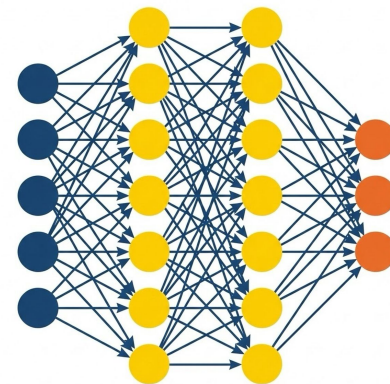
Step-by-step tutorial demonstrating how to annotate cell types using Azimuth (a reference-based annotation tool), convert the results into a .cloupe file using LoupeR, and explore the results in Loupe Browser.

Automated cell annotation models in Cell Ranger



10x Cloud-based

- Supported species: human and mouse
- Internet access: required
- Model type: Nearest neighbor
- Model co-developed with Broad Institute



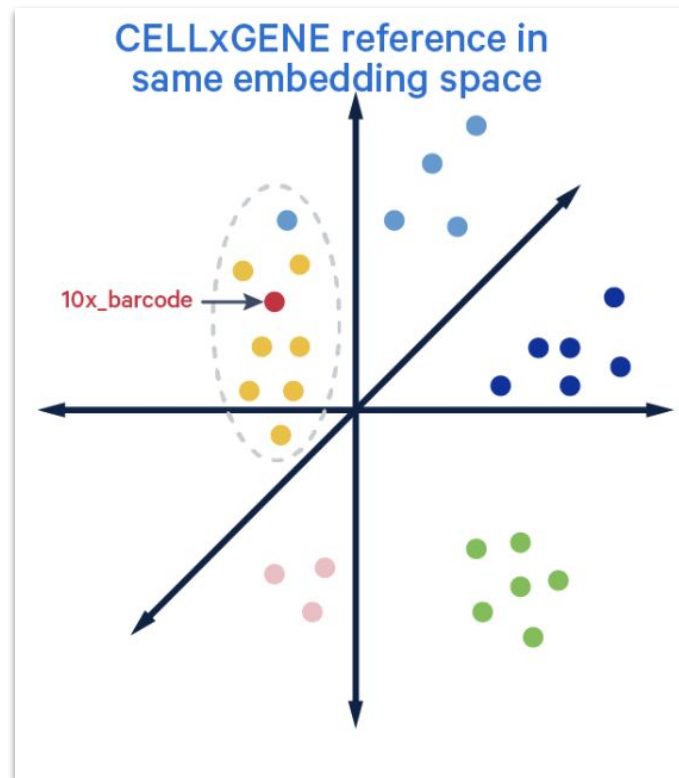
Pan-Human Azimuth

- Supported species: human only
- Internet access: not required
- Model type: Neural network classifier
- Developed by Satija Lab

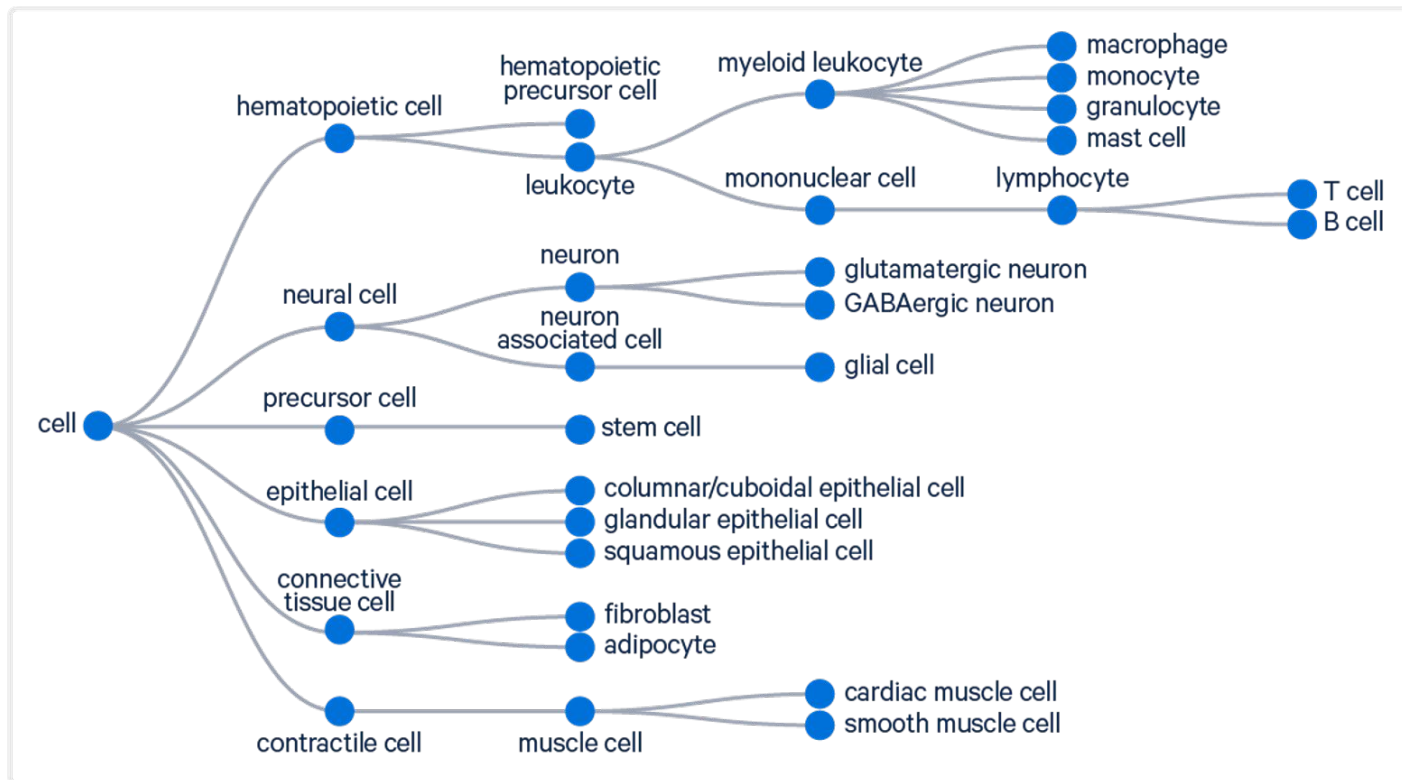
Cellranger 10x Cloud-based annotation

A starting point for your annotation journey

Each cell's gene expression profile is compared to a model built on the Chan Zuckerberg CELL by GENE census to identify the most similar cell types. A consensus label is then assigned to each barcode.



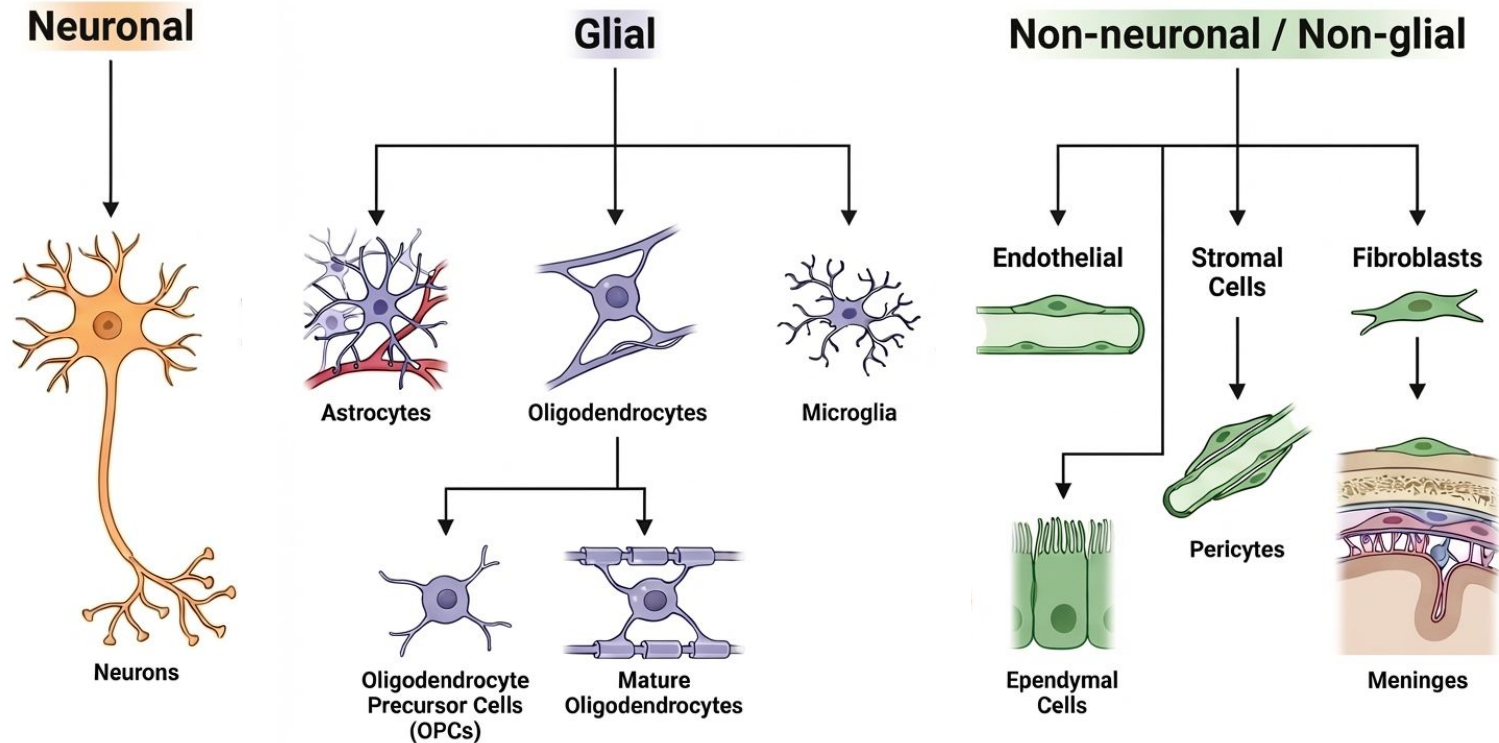
How it works: cell ontology



Example from the 10x Cloud-based models, annotations terms are from Cell Ontology.

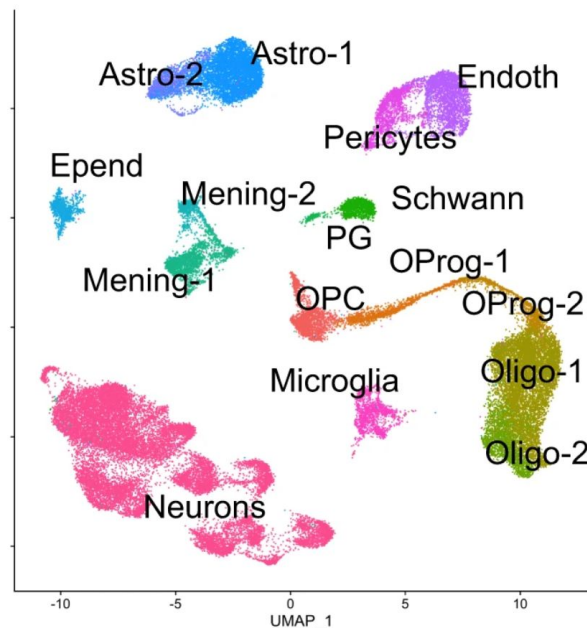
Major cell types of the nervous system

Neurobiology 101

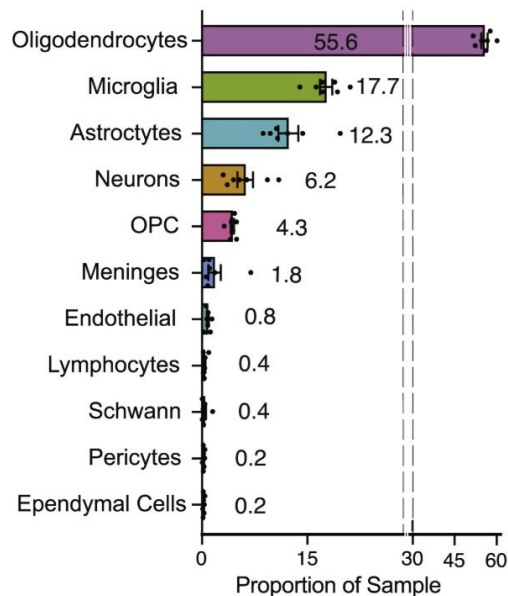


What cell types do we expect in our data?

Go to the literature to identify likely cell types



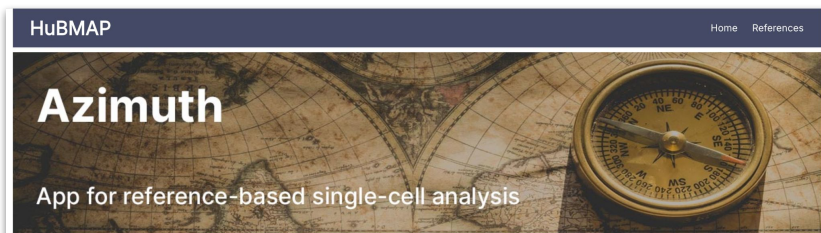
<https://doi.org/10.1038/s41467-021-25125-1>



<https://doi.org/10.1016/j.neuron.2023.01.007>

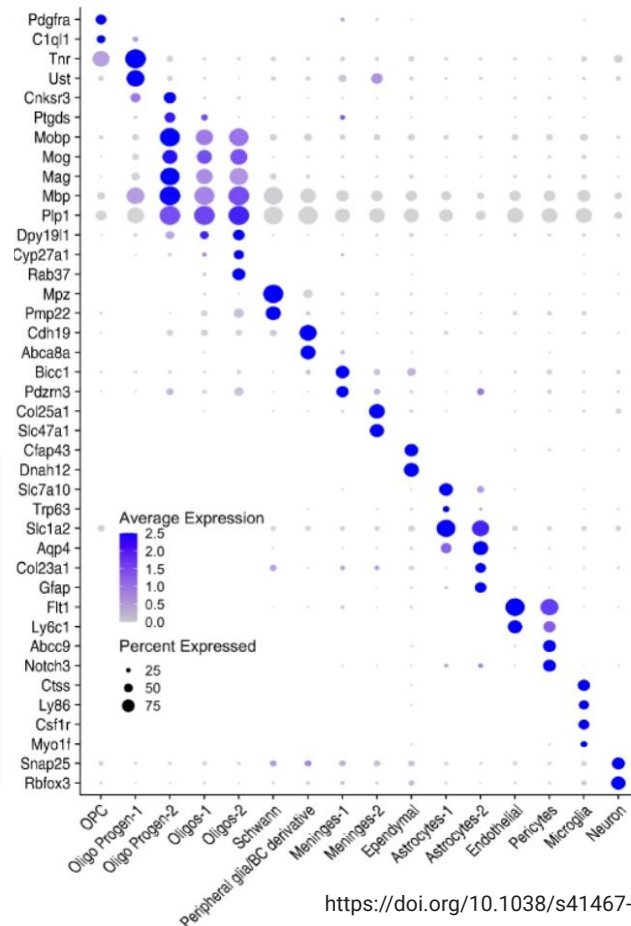
Canonical markers

Use information from the literature to put together cell type marker list



Label	Expanded Label	OBO Ontology ID	Markers
Astro_1	Astrocyte_1	astrocyte	Gfap, Aqp4, Myoc, Pla2g7, Igfbp5, Id4, Mlc1, Ntsr2, Sdc4, Gja1
Astro_2	Astrocyte_2	astrocyte	Slc1a2, Gpc5, Wdr17, Htra1, Slc1a3, Acsf3, Plpp3, Msi2, Dtna, Gabrb1
Chandelier	Chandelier neuron	GABAergic neuron	Pthlh, Pde3a, Mctp2, Kit, Unc5b, Epb41l4a, Mme, 4933432K03Rik, Gm44129, A830036E02Rik
Endo	Endothelial cell	endothelial cell	Flt1, Sgms1, Adgrl4, Slc2a1, Slco1a4, Bsg, Klf2, Ptxr, Sotbn1, Mecom

[Azimuth Hubmap](#)

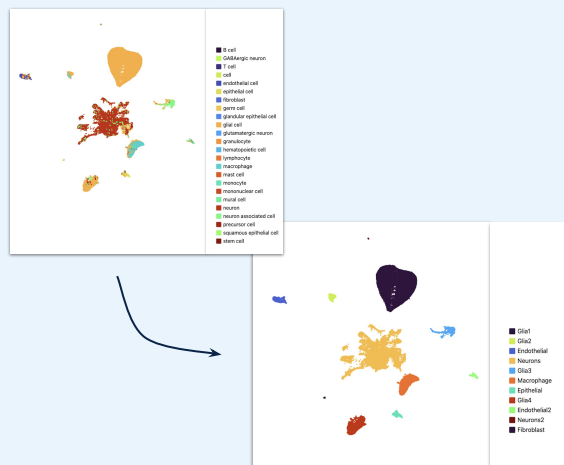


<https://doi.org/10.1038/s41467-021-25125-1>

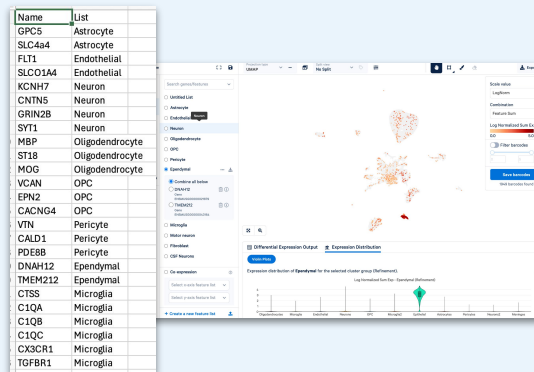
Loupe annotation overview

Main steps and data in Loupe annotation demo

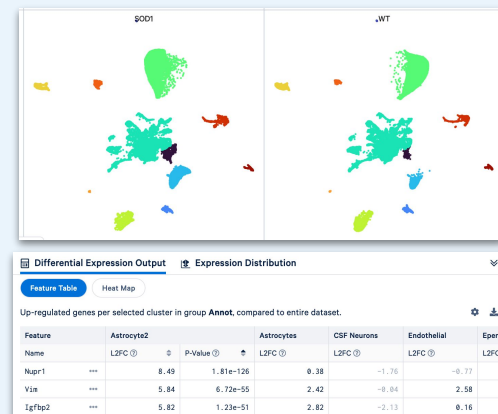
Initial annotation using Cellranger Annotate results

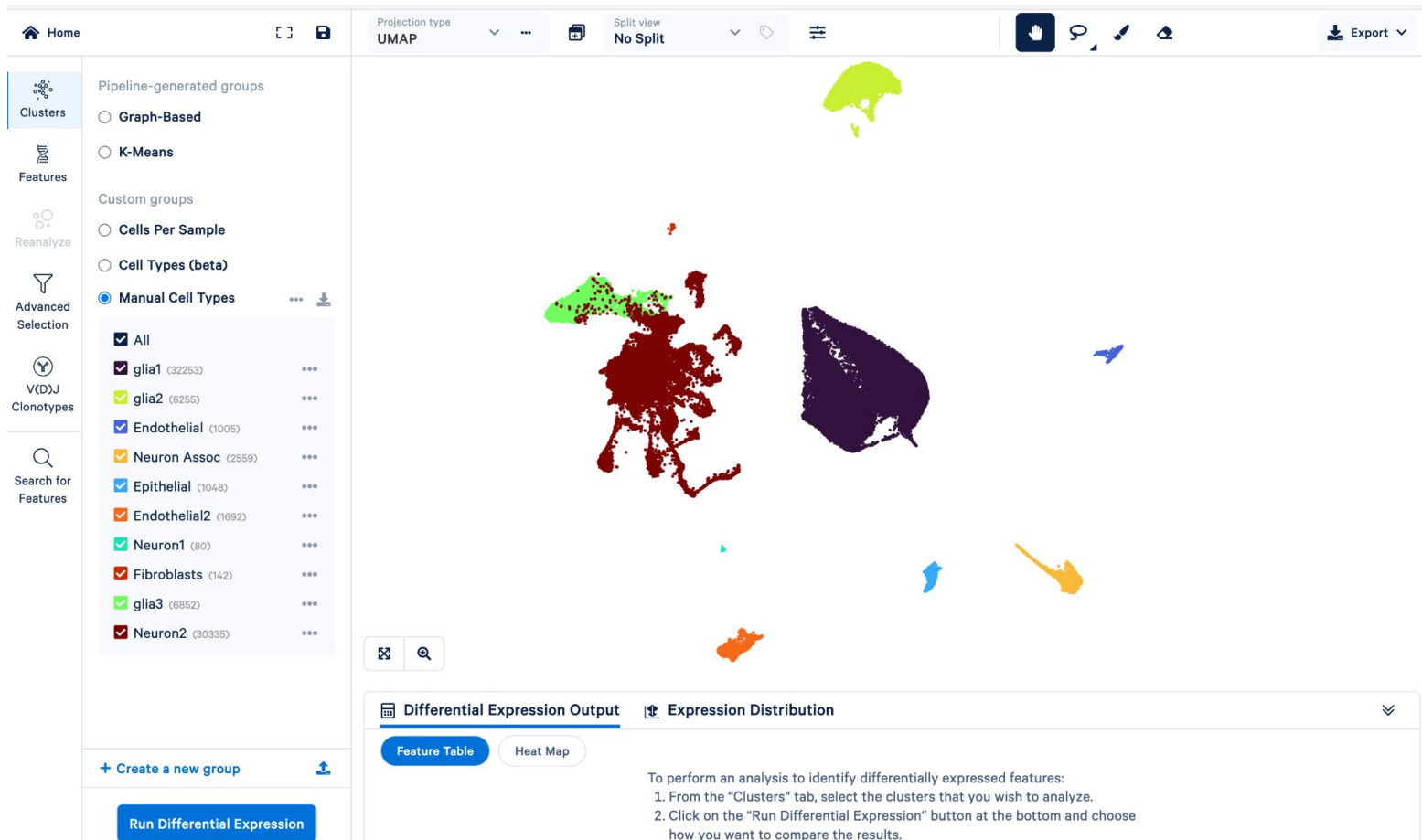


Annotation refinement with canonical marker genes



Exploration of data with cell types assigned





Interactive sessions

Glia3 cluster numbers:

6

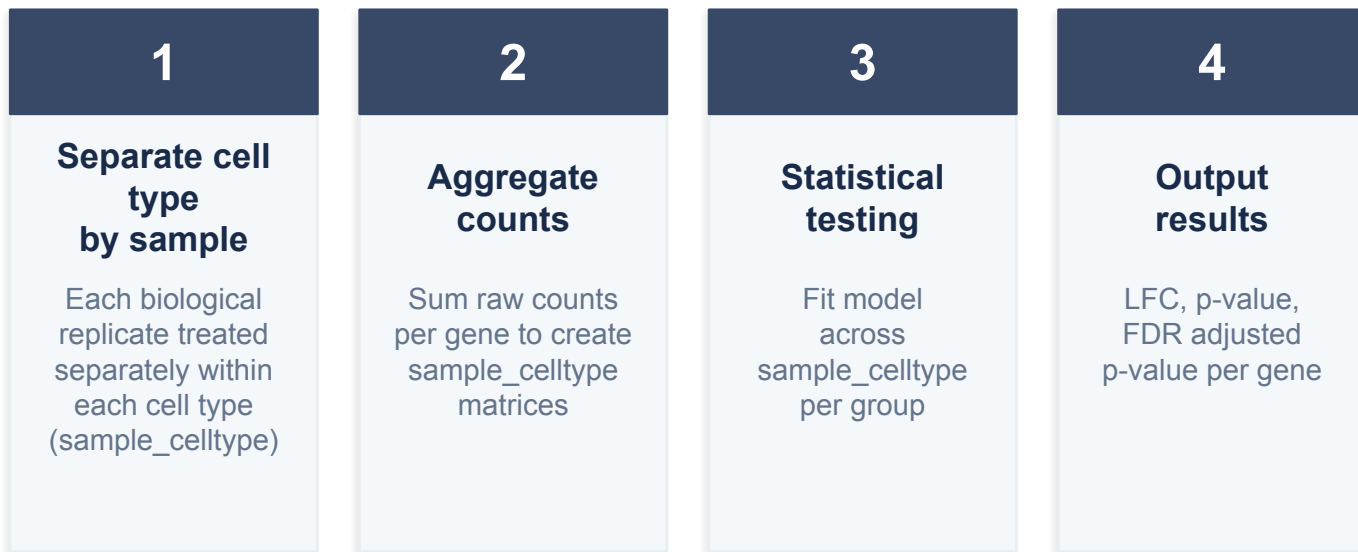
10

18

33

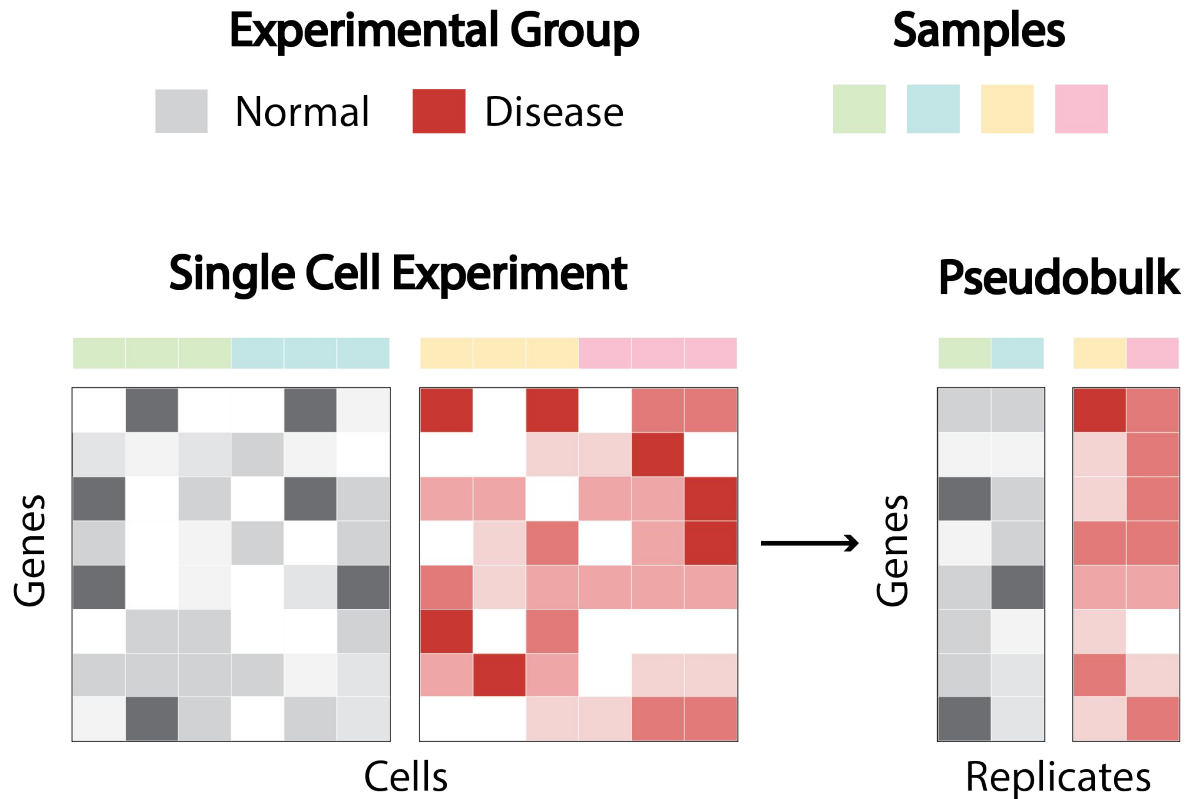
41

The pseudobulk approach (recommended)



Key insight: pseudobulk treats each sample as one observation — correctly propagating biological variability across replicates

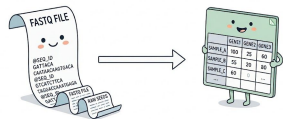
Pseudobulk approach to DGE



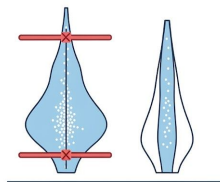
Interactive sessions

Common analysis flow for single cell data

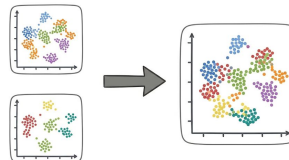
Cell Ranger:
read processing &
secondary
analysis



QC & data filtering



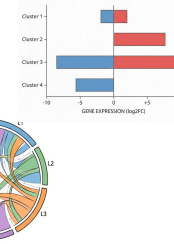
Data integration



Visualization and
exploration



Multisample
comparison and
more



Seurat object qs2 file

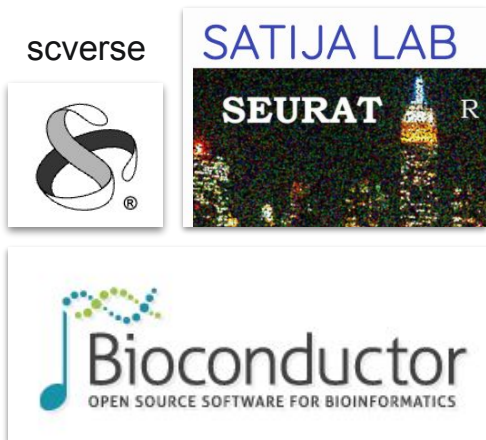
Introduction to using community developed tools

Hands on tour of the single cell analysis journey

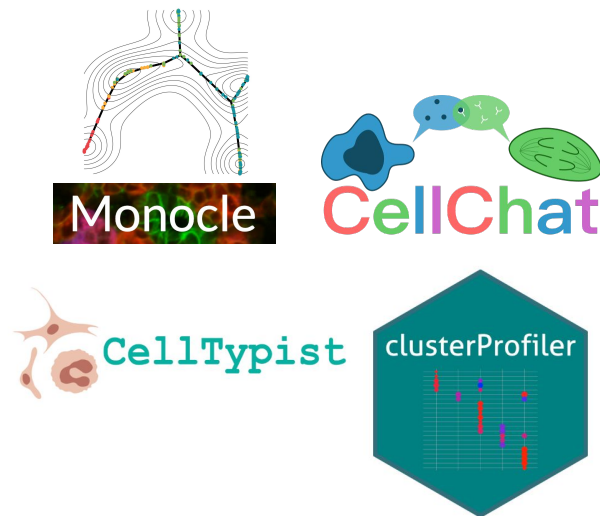
Outline

- Approaching community developed tools
- Specialized tools for single cell analysis
- Using AI for single cell analysis
- Additional resources

Community developed tools



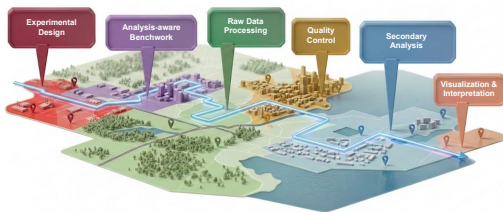
Ecosystems



Specialized Analysis Tools

How to choose which tool to use

A workflow-guided approach for researchers new to single cell analysis



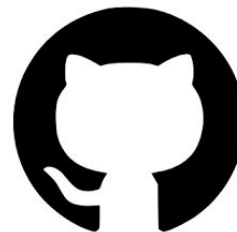
Workflow guides tool selection

- What analysis helps you answer your research question?
- You should only look for a tool when you hit a problem.



Follow benchmarks and curated resources

- Benchmarking studies offer best-practice guidelines.
- You can consult [Analysis Guides](#) published by 10x Genomics.



Look at the code repository

- Has the GitHub been updated recently?
- Do the developers respond to issues from the community?

How to learn community tools

Finding tutorials and posting issues

Who should I ask if I have questions about community tools?

README GPL-3.0 license

- Installation on Mac OS with R > 3.6: Please re-install [Xquartz](#).
- Installation on Windows, Linux and CentOS: Please check the solution for [Windows](#) and [Linux](#).

Tutorials

Please check the tutorial directory of the repo. Example datasets are publicly available at [figshare](#). Please check the [Jin et al., Nature Protocols 2024](#) for a comprehensive protocol of cell-cell communication analysis using CellChat.

Analysis of single-cell transcriptomics data

- [Full tutorial for CellChat analysis of a single dataset with detailed explanation of each function](#)
- [Full tutorial for comparison analysis of multiple datasets](#)
- [Comparison analysis of multiple datasets with different cellular compositions](#)

Analysis of spatially resolved omics data

- [Brief tutorial for CellChat analysis of a single spatially resolved transcriptomic dataset](#)
- [Brief tutorial for CellChat analysis of multiple spatially resolved transcriptomic datasets](#)
- [Brief tutorial for CellChat analysis of spatial multiomics data](#)
- [Frequently Asked Questions when analyzing spatially resolved transcriptomics datasets](#)

Additional utilities

- [Interface with other single-cell analysis toolkits \(e.g., Seurat, SingleCellExperiment, Scanpy\)](#)
- [Tutorial for updating ligand-receptor database CellChatDB](#)

jnworks / CellChat

< Code Issues 337 Pull requests 14 Actions Projects Security Insights

You only have a single verified email address. We recommend verifying at least one more email address to ensure you can recover your account if you lose access to your primary email.

Save netVisual_circle plot as PDF #329

New Issue

Closed

erikagucciardo opened on Mar 20, 2025

Hi! Thank you for this great tool!

I was trying to save the "netVisual_circle" plots as PDF, but without success.

Here the code I use:

```
pdf(file = paste("Circle_plot_interactions", ".pdf", sep=""))
netVisual_circle(cellchat@net$count, vertex.weight = groupSize, weight.scale = T, label.edge = F, title.name = "A", graphics.off())
```

Am I missing something?

Thanks for any help!

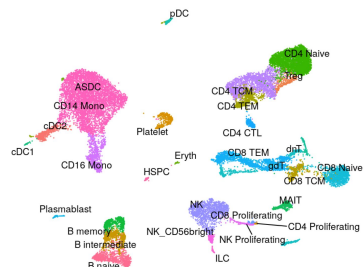
suye0620 on Mar 21, 2025

Like this:

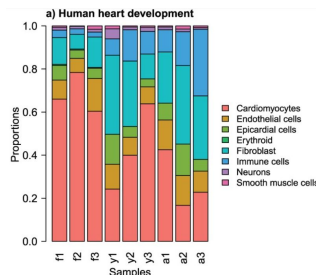
```
for(LR in ChatLR$LRsigInteraction_name){
  tryCatch(
    cat(LR)
    # pdf(file = paste("myfiles/test_plot.pdf"))
    png(string = str_c(".", ChatPlots2/circlePlot_",LR_", ".png"))
    par(mfrow=c(1,1), xpd = T) # "xpd = TRUE" should be added to show the title
```

Specialized analysis tools developed by community

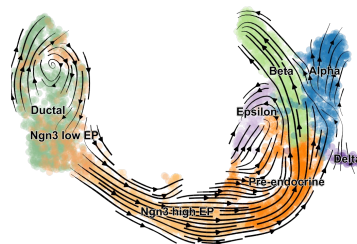
Most require programming skills (R or Python)



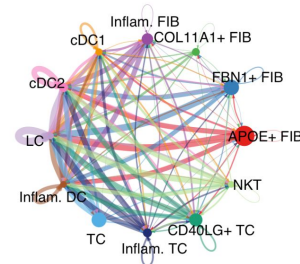
Automated cell-type
annotation



Cell type
composition analysis



Trajectory inference



Cell-cell interaction
analysis

[Cell annotation](#)

[Introduction](#)

Web Resources for Cell Type Annotation

[Cell ordering](#)

[Tutorial](#)

Mar 22,

**Trajectory Analysis using 10x Genomics Single
Cell Gene Expression Data**

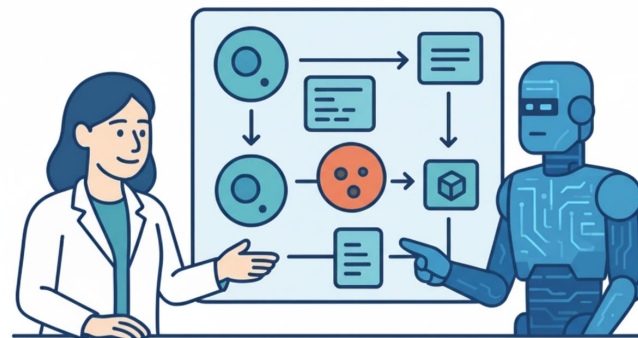
What other
analyses have
you heard of?

Using AI for Single-Cell Analysis

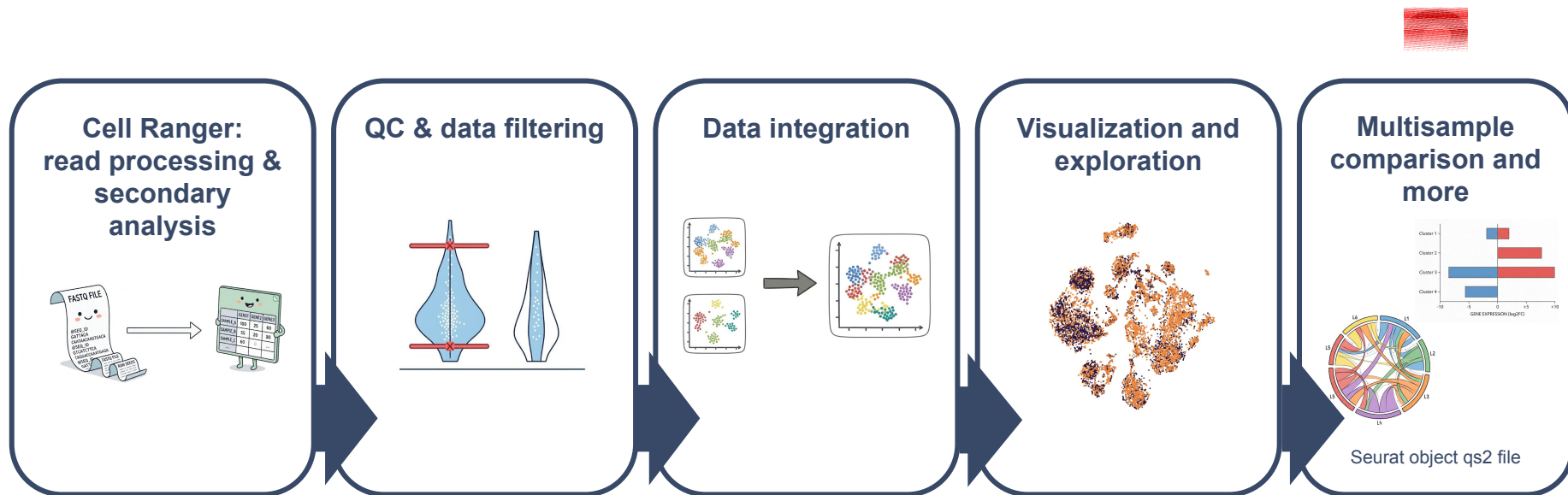
AI as a Scientific Tool



AI as a Coding Partner



Common analysis flow for single cell data



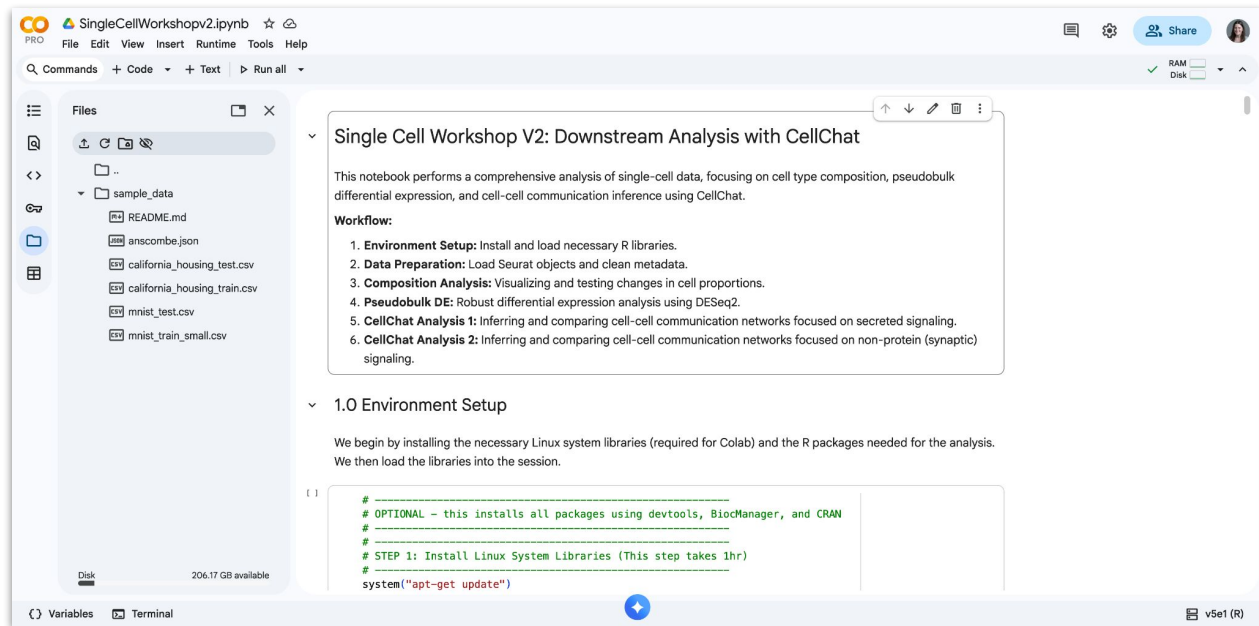
Further analysis using community developed tools

Hands on tour of the single cell analysis journey

Outline

- Next steps of downstream analysis
- Hands-on Demo: preparing environment and data
- Slides: Seurat object
- Hands-on Demo: Analysis with Seurat
- Slides: cell-cell interaction analysis
- Hands-on Demo: CellChat

Quick introduction to .ipynb and Colab

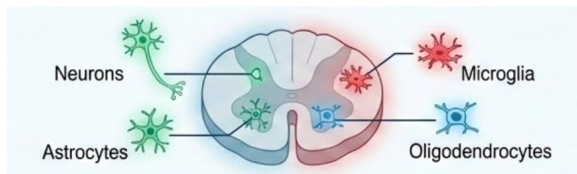


Purpose of the experiment

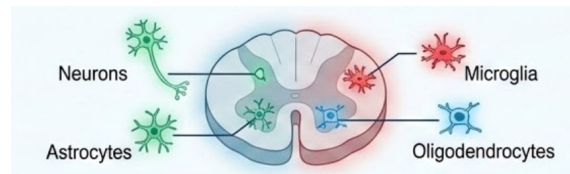
Our research questions that will drive the analysis:



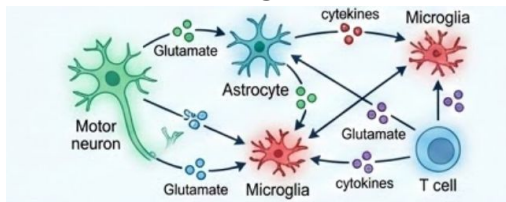
What cell types exist in our sample?



How do cell types change in ALS?



How do cell-cell interactions change in ALS?



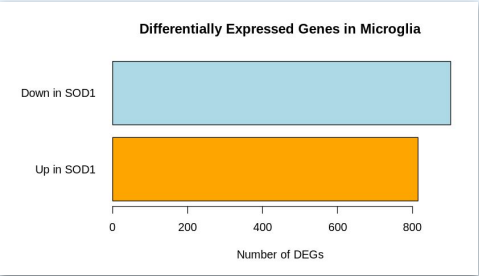
Can we identify novel therapeutic targets with our results?



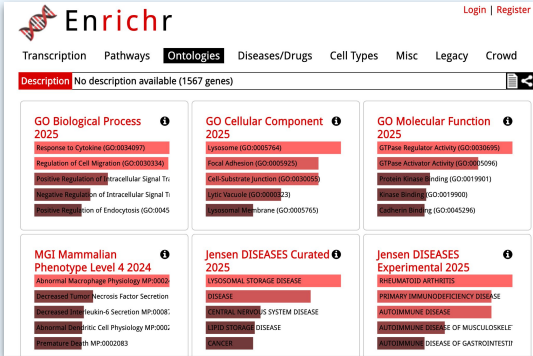
Downstream analysis workflow

Main analyses we will perform in the notebook

Differential gene expression



Enrichment of DEGs



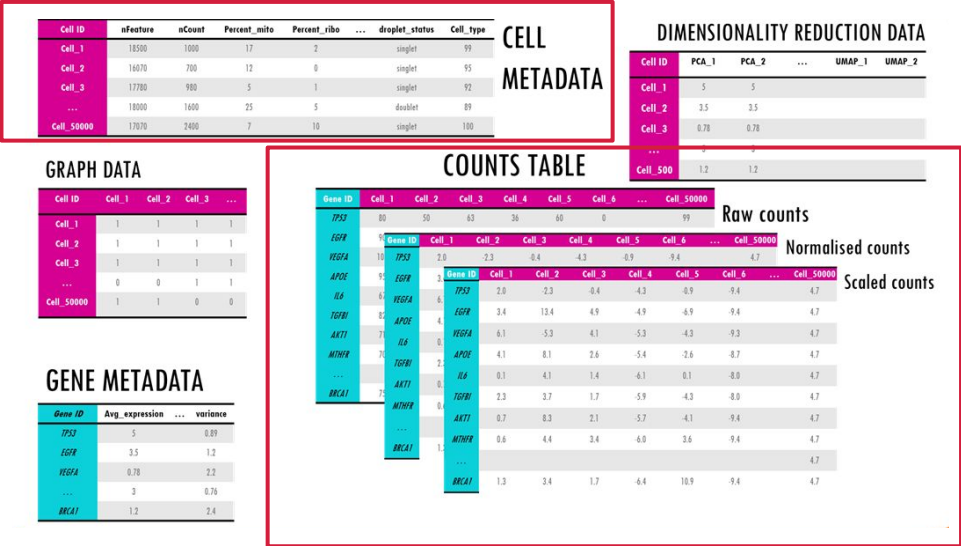
Cell-cell interaction

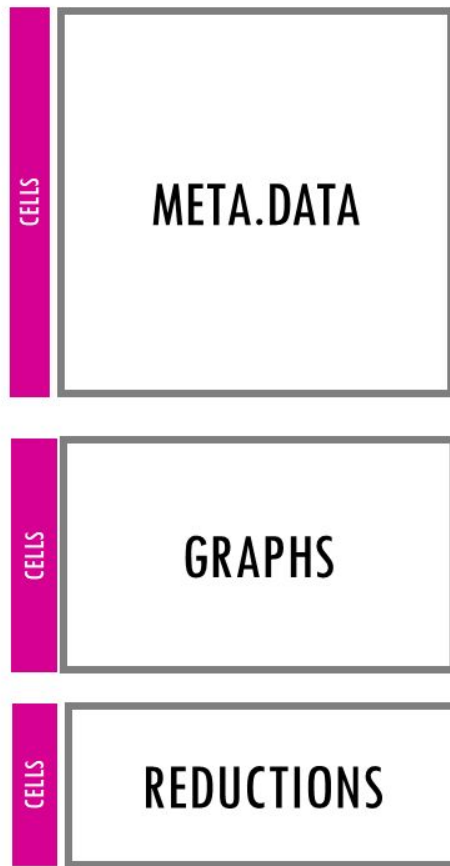


Interactive session

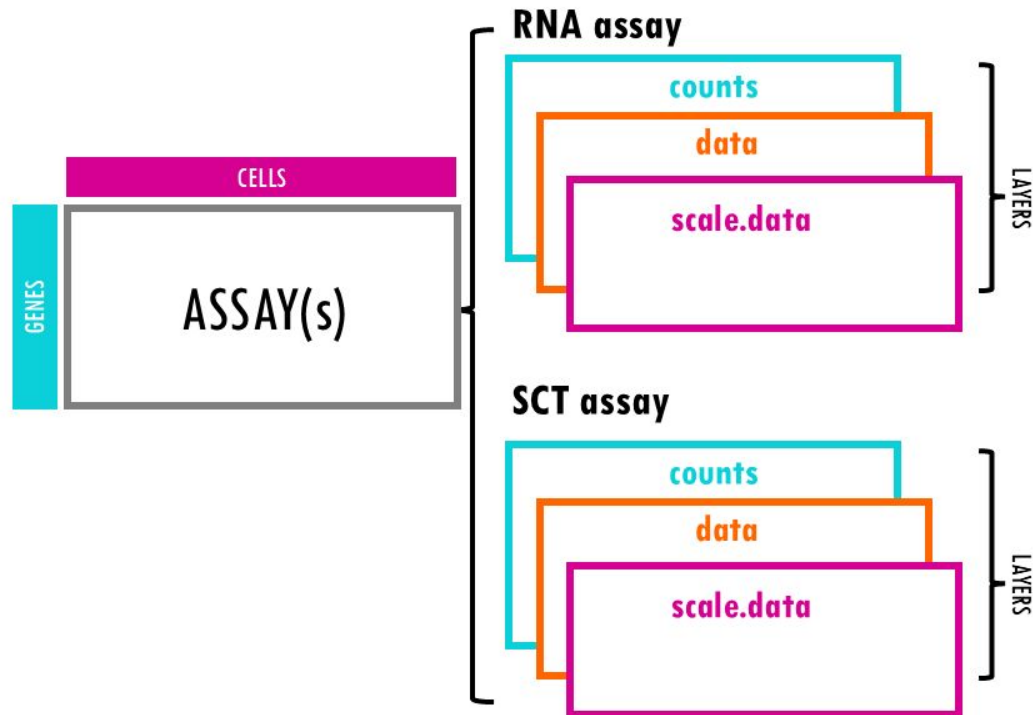
Seurat object is an organized container for single cell data

Single Cell Data





SEURAT OBJECT

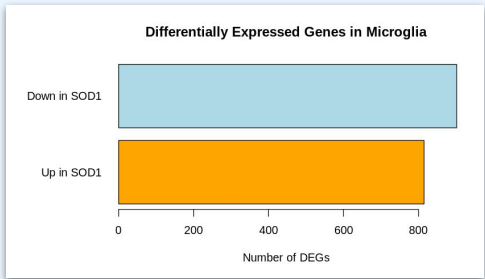


Downstream analysis workflow

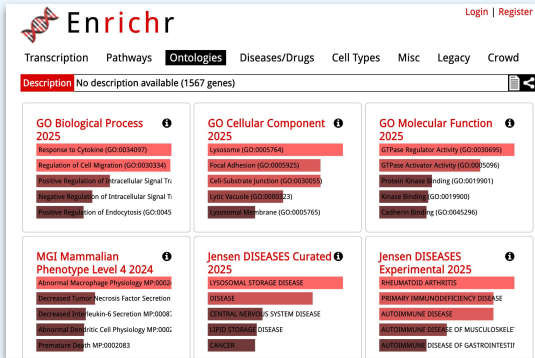
Main analyses we will perform in the notebook



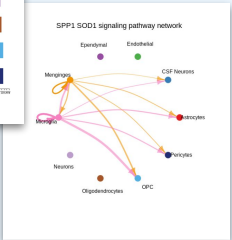
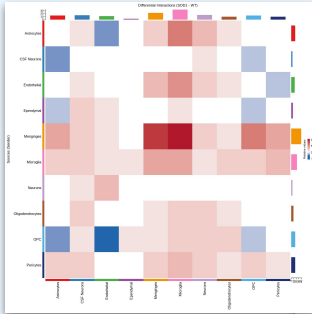
Differential gene expression



Enrichment of DEGs



Cell-cell interaction

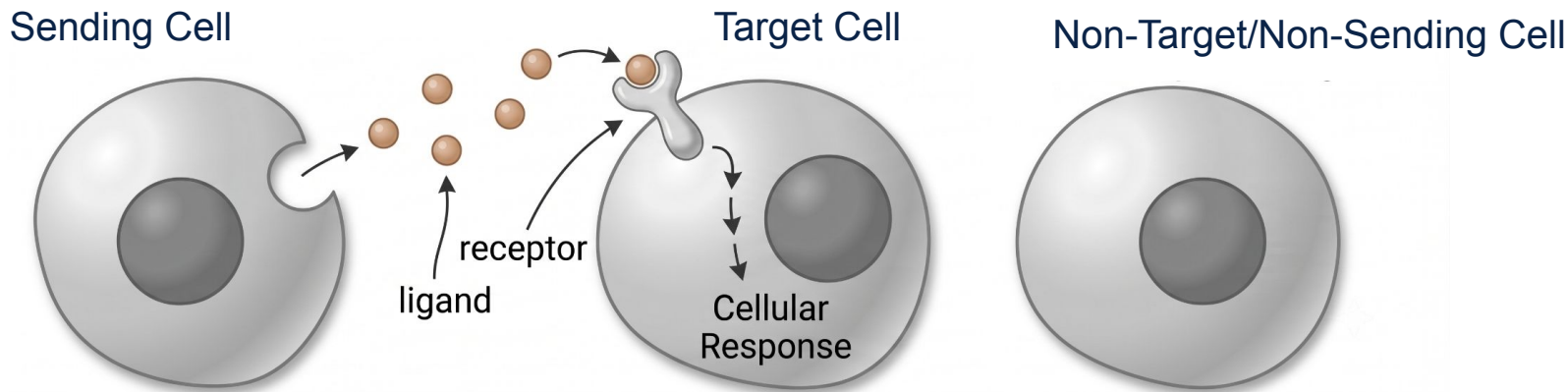


Interactive session

Introduction to Cell-Cell Interaction analysis

What are cell-cell interactions?

Ligand-receptor interactions inform cell and tissue function



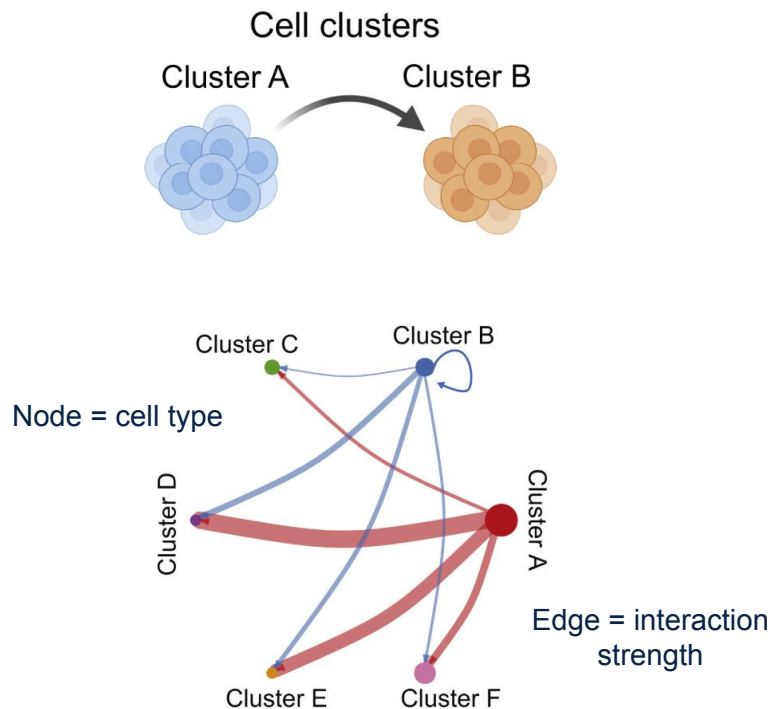
Example questions:

- What signaling pathways are altered in disease states?
- How do cells direct differentiation during development?
- What communication pathways are critical drug targets?

106

How do we map cell-cell interaction using single cell data?

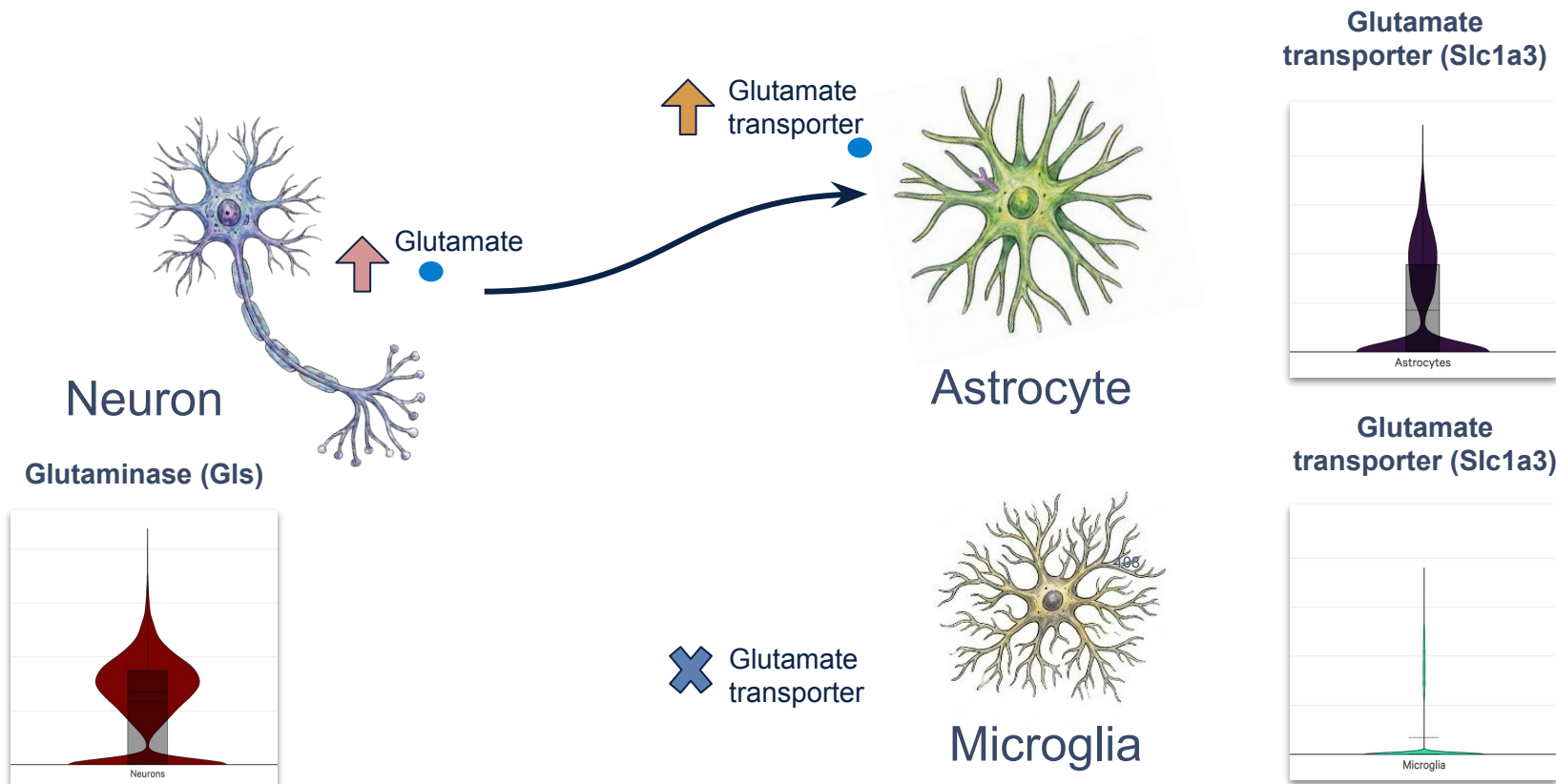
- 1 Measure ligand and receptor expression from each cell-type gene profile.
- 2 Match ligands to receptors using a database of known interactions to map cellular communication.
- 3 Infer communication patterns using expression + database information.



<https://doi.org/10.1016/j.coisb.2021.03.007>

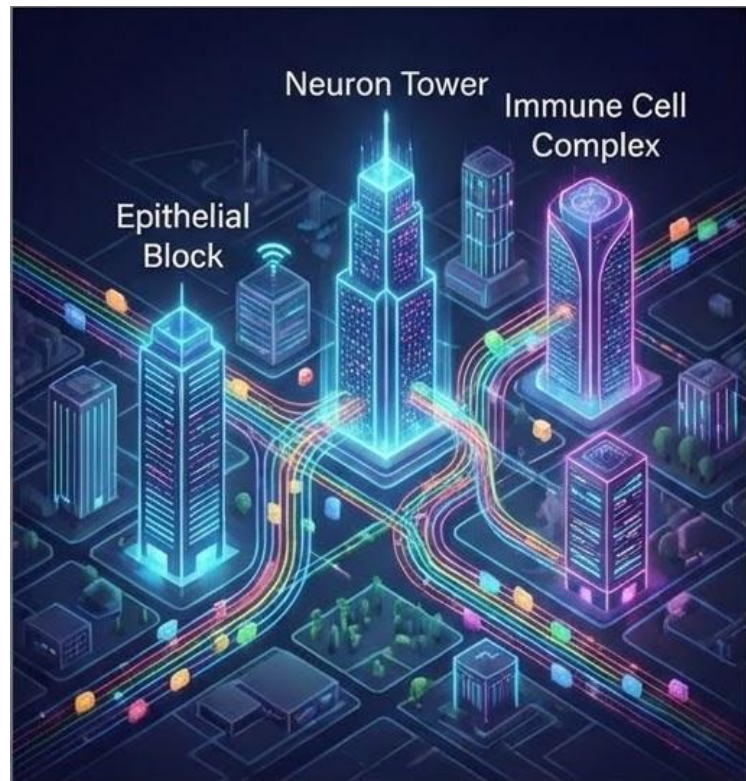
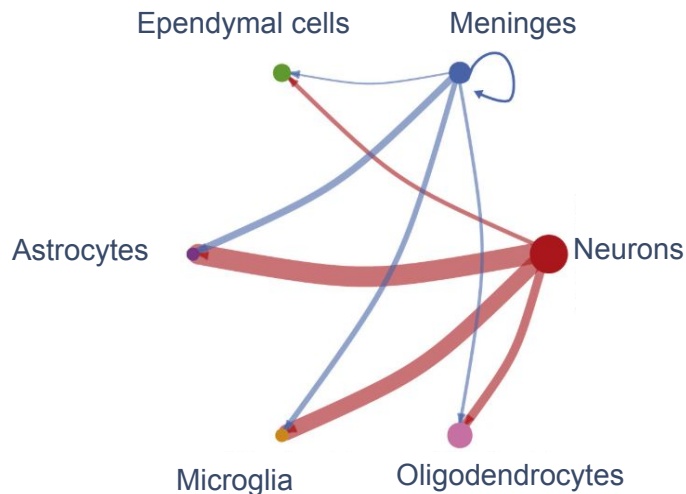
Example: using gene expression to infer glutamate signaling

Neurons release glutamate; astrocytes take up glutamate



What does this type of analysis tell us?

Cell-cell interaction analysis tells us who is talking to whom **and** what they are saying.



Cell-cell interaction demonstration

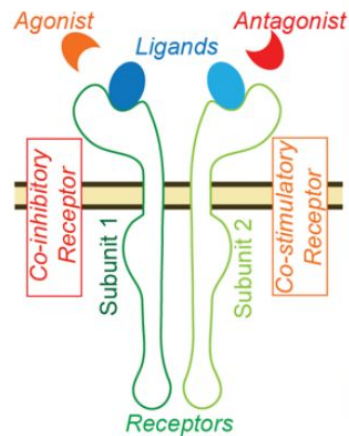
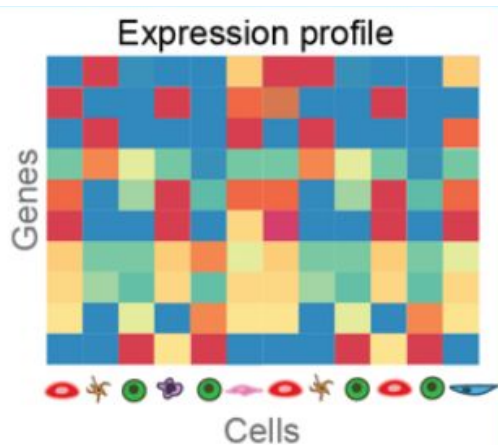


Tool: **CellChat**

An R package designed to infer, visualize, and analyze cell-cell communication networks

Inputs

Single cell data, CellChat database



CellChatDB

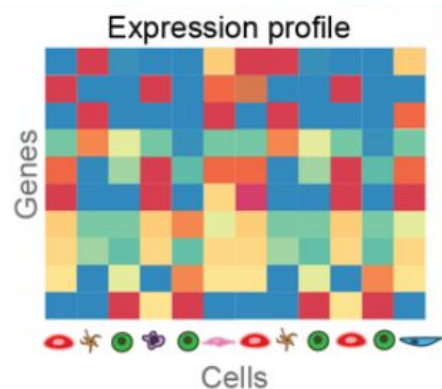
(~3,300 L-R pairs)



How are cell-cell interactions determined?

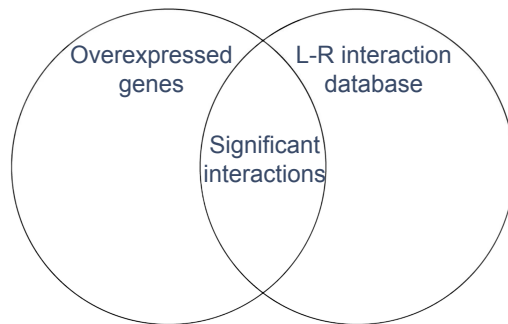
1

Identify overexpressed genes in each cell type



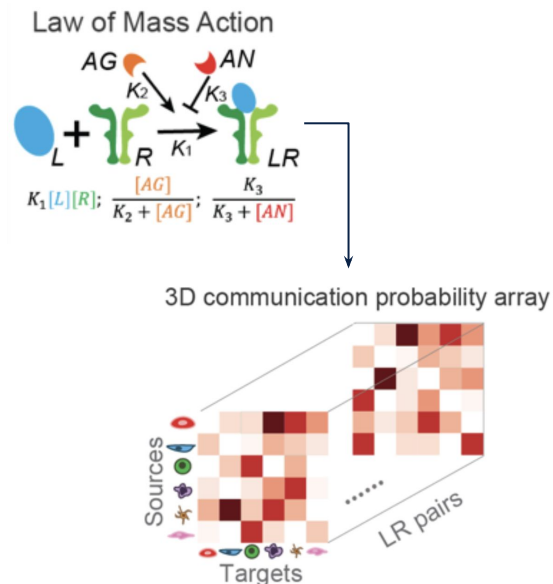
2

Identify overexpressed interactions



3

Calculate communication probability between cell types



How are cell-cell interactions determined? (continued)

4

“Pathway analysis” of the significant interactions

CXCL

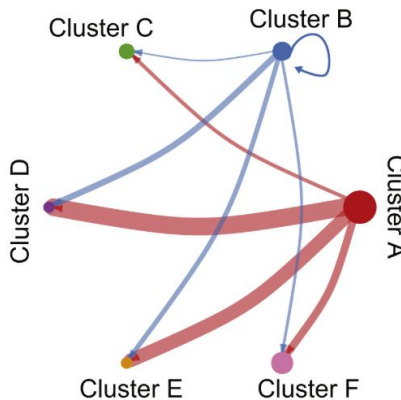
Cxcl12 - Cxcr4
Cxcl12 - Cxcr7
Cxcl1 - Cxcr2

WNT

Wnt1 - FZD1
Wnt1 - FZD8
Wnt5a - FZ4

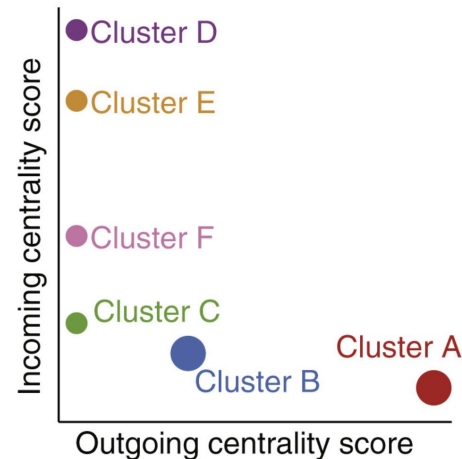
5

Summarize the aggregated network



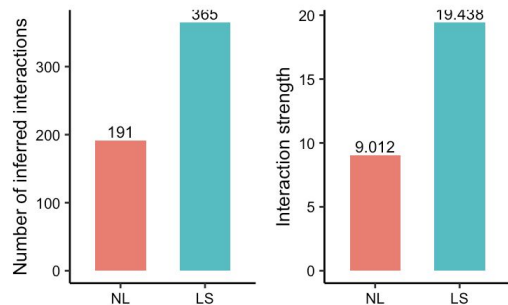
6

Identify senders and receivers

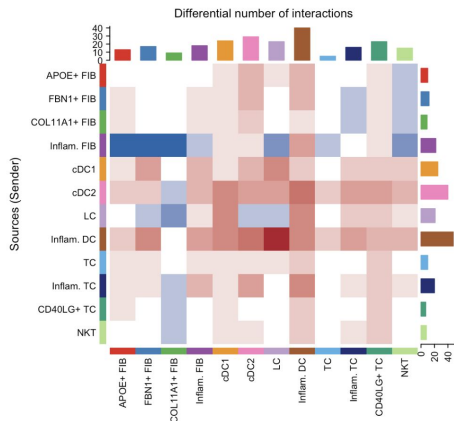


An example with a normal skin vs diseased dataset

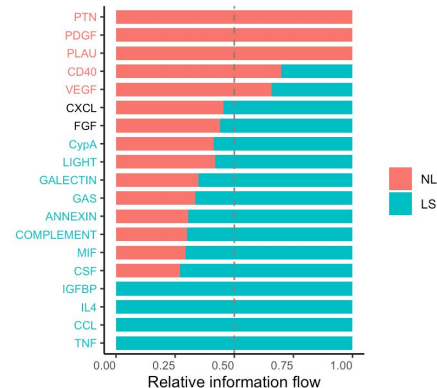
What questions are the following plots answering?



What are the number and strength of interactions?



How do cell interactions change in the disease condition?



What are the altered pathways in the disease vs. normal condition?

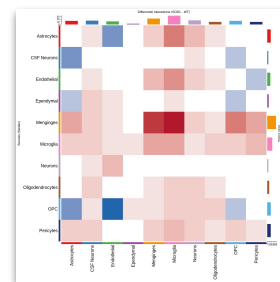
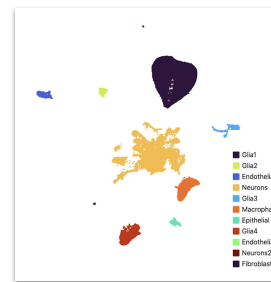
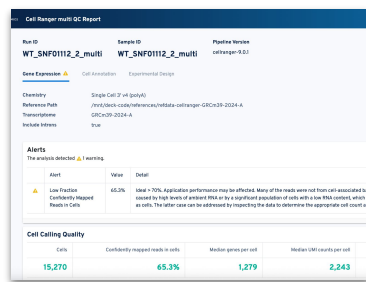
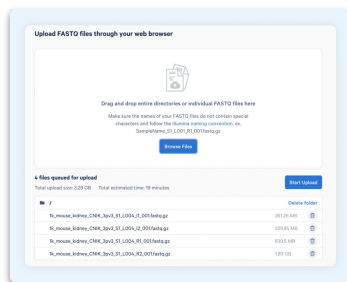
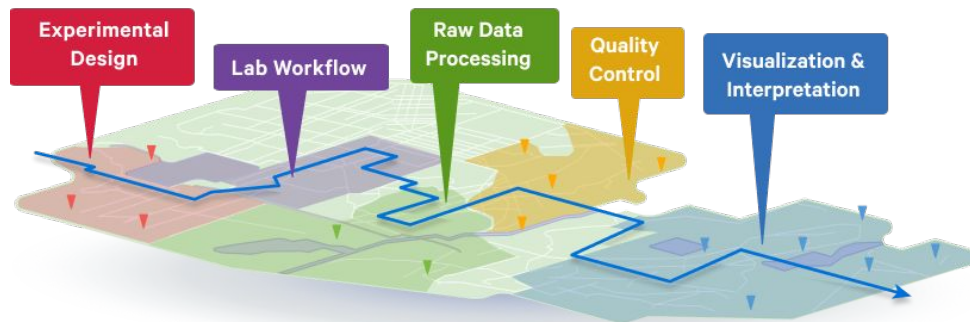
Interactive session

Quick Recap

Hands on tour of the single cell analysis journey

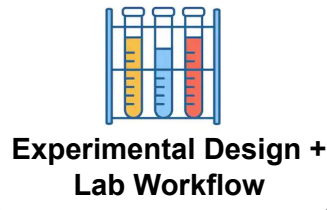
Recap

Our journey through analysis

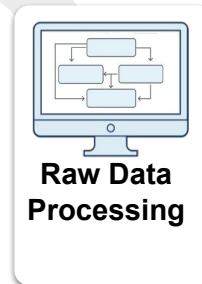


We are here to support you!

- 10x customer publications
- 10x-pert & customer webinars
- Open-source community tools
- 3rd party analysis services and products



- Discussion with local Field Application Specialist and Science and Technology Advisor



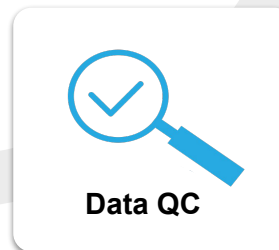
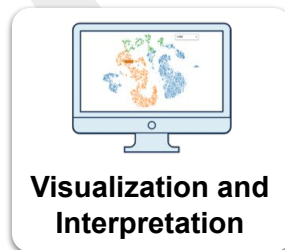
- Software training from FAS
- Support site documentation and software tutorials



ASK US!

support@10xgenomics.com

- Support site documentation and software tutorials



- Web Summary
- Data QC Technical Note and Analysis Guide

Early Access Opportunity: 10x Genomics Analysis Agent

- 10x Genomics is building a new software platform to analyze Chromium data by asking questions in plain English.
- We're inviting a limited number of researchers for early access and direct input on the roadmap.
- Scan the QR code or visit the link to request access.



Sign up for the
10x Genomics Analysis Agent Beta:
10xgen.com/genetic-analysis-beta



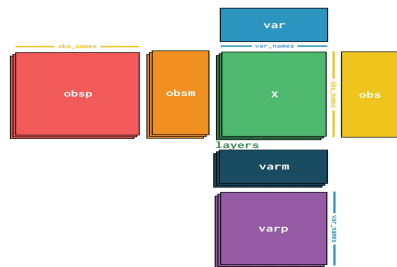
10x Genomics Data Analysis Course

A special announcement

10x Data Analysis Course: A comprehensive guide to single cell and spatial transcriptomics

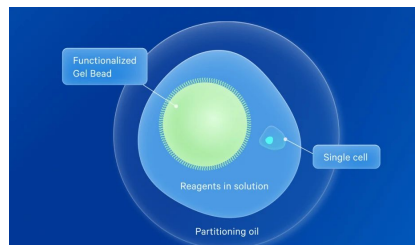
8-week cohort-based online course

Phase 1: Foundations



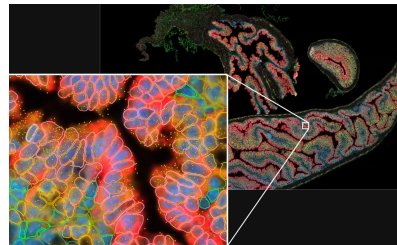
Bioinformatician's tool kit:
Linux, Python, GitHub,
Experimental design

Phase 2: Single cell



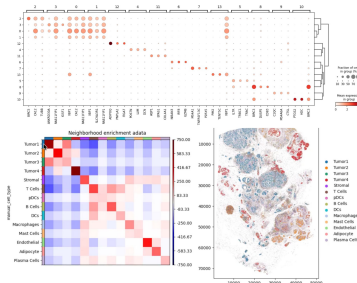
The cell is the basic unit of
life: data processing, QC,
annotation, community
developed tools

Phase 3: Spatial



Context matters: decoding,
segmentation,
spatially-aware analysis

Phase 4: Synthesis



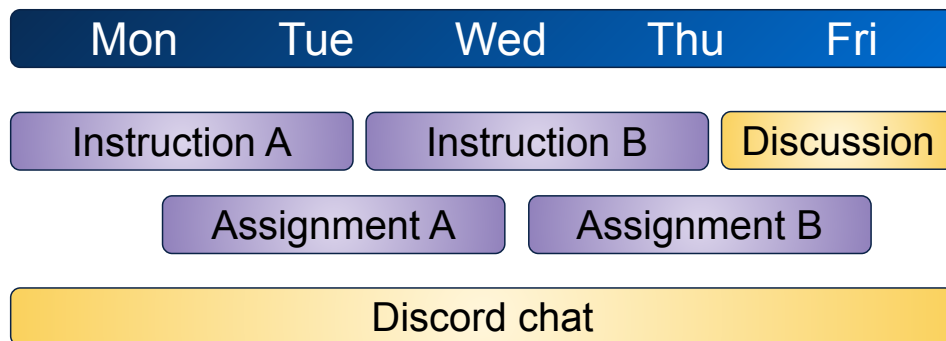
Bringing it all together:
publication quality figures,
keeping current,
communicating results

AI accelerators for every step of the journey

10x Data Analysis Course: A comprehensive guide to single cell and spatial transcriptomics

- Online cohort-based course
 - Pre-recorded instructions
 - Assignments
 - Weekly live discussion
 - Group Discord channel for interacting with instructors and fellow students
- ~ 8 hours/week student commitment
- Cohort size of 40-60 students
- Certificate of completion
- Post-course Alumni discord server to maintain relationships

Course format for a typical week



Pre recorded



Live

10x Data Analysis Course: A comprehensive guide to single cell and spatial transcriptomics

- Who is this course for?
 - Researchers and staff across academia and industry seeking to invest in themselves and gain independence in analysis

General

Workshop

Jul 27, 2026

Level Up Your Research: The 10x Data Analysis Course

The journey from experimental design to publication-ready figures is one of the most challenging aspects of modern biology. To help biologists overcome the steep learning curve of bioinformatics, 10x Genomics is introducing a comprehensive, 8-week online data analysis course covering single cell and spatial transcriptomics.

<https://www.10xgenomics.com/analysis-guides/level-up-your-research-10x-data-analysis-course>

Application Form

